

**Household Framing in Bank of England
and Press Communication:
Evidence from UK Inflation Expectations**

Timur Kayaalp

Dissertation Supervisor: Professor Stephen Hansen

Dissertation submitted in part-fulfilment of the MSc in Economics,

University College London

September 2026

*Dedicated to all political prisoners, to those who have lost their freedom
for daring to speak, to dissent, and to work for the freedom of others.*

Declaration

I, Timur Kayaalp, hereby declare that the work presented in this dissertation is my own original work. Where information has been derived from other sources, I confirm that this has been clearly and fully identified and acknowledged. No part of this dissertation contains material previously submitted to the examiners of this or any other university, or any material previously submitted for any other assessment.

Name: Timur Kayaalp

Date: 2 September 2026

Classification

This piece of research is primarily (please indicate as appropriate):

- ☒ an empirical/econometric study
- ☐ the examination of a theoretical problem
- ☐ a critical analysis of a policy issue
- ☐ an analytical survey of empirical and/or theoretical literature

Dissertation Library

I give permission for a pdf electronic copy of this dissertation to be made available to future students by way of the Dissertation Library, should the Department of Economics choose to use it in this way.

Yes / No

please indicate your preference

Table of Contents

1	Introduction	1
2	Literature and contribution	3
2.1	Communication and household expectations	3
2.2	Measuring text	4
2.3	Identification	5
3	Data	7
3.1	Household inflation expectations	7
3.2	Bank of England text	7
3.3	Newspaper text	8
3.4	Macroeconomic, market and policy series	9
4	Measurement	11
4.1	A dictionary measure of stance	11
4.2	Classifier measures	12
4.3	Press household framing	12
4.4	Validation	13
5	Empirical strategy	15
5.1	Constructing the surprise	15
5.2	Local projections	16
5.3	Inference	16
5.4	Identification	17
6	Results	20
6.1	The audience channel	20
6.2	Is the result an artefact of the word list?	21
6.3	Does communication carry information?	23
6.4	When does the channel operate?	25
6.5	Separating stance from information	26
6.6	The distribution of household beliefs	28
7	Discussion	32
7.1	The results in light of Carroll (2003)	32
7.2	Relation to Tong and Cherry (2026)	33
7.3	Relation to other work	34
8	Conclusion	35
	Bibliography	37
	Appendix 1: Dictionaries and word lists	40
	Appendix 2: Variables and data construction	42
	Appendix 3: Additional results	47

List of Figures

1	Impulse responses to a household-framing surprise	20
2	Keyword and classifier measures compared	22
3	Rotated impulse responses	27
4	Household framing and the belief distribution	29

List of Tables

1	Empirical size of the local projection under two estimators	17
2	Household framing and one-year household inflation expectations	20
3	Text surprises as predictors of macroeconomic and market outcomes	23
4	Press household framing and the implied inflation curve, by maturity	24
5	Publication-day response of the implied inflation curve, pre-2015 minutes	25
6	The framing effect at different levels of press attention	25
7	Stance and information shocks, entered jointly	27
8	Press household framing and the Ipsos belief distribution	28
9	Identified shocks against the belief distribution	30
10	Candidates removed from the hawkishness dictionary	41
11	Household inflation expectations	42
12	Macroeconomic series	42
13	Market series	43
14	Text and attention measures	43
15	Policy and forecast series	43
16	Newspaper corpus construction, 172 months	44
17	Stage 1 — first-stage fit and the share of tone that is a surprise .	47
18	Stage 1 — first-stage coefficients, cb_tone	47
19	Stage 2 — validation	47
20	Stage 3 — episode comparison, dictionary	48
21	Stage 3 — episode comparison: cb_tone	48
22	Stage 4 — local projections (Jordà 2005): hawk_boe , full sample .	48
23	Stage 4 — local projections (Jordà 2005): hawk_boe , moving regime	49
24	Stage 4 — local projections (Jordà 2005): cb_tone , full sample . .	49
25	Stage 4 — local projections (Jordà 2005): cb_tone , moving regime	50
26	Stage 5 — the placebo fails: the surprise ‘predicts’ expectations before it occurred	50

27	Stage 5 — fix (a): controlling for the six-month pre-trend. The effect survives	50
28	Stage 5 — fix (b): first-differencing the surprise. The effect vanishes	50
29	Stage 5 — quiet-regime placebo, where expectations are flat by construction	51
30	Stage 6 — robustness of the headline specification ($h = 10$, moving regime)	51
31	Stage 7 — mechanism (Carroll’s second implication)	51
32	Stage 9 — market validation	52
33	Part A — lead placebos on the headline specification: Press household framing	52
34	Part A — lead placebos on the headline specification: BoE household framing	52
35	Agreement between the two series	53
36	The same impulse response, both measures	53
37	GFC (2007–01 to 2009–12)	53
38	COVID+ (2020–01 to 2026–07)	54
39	COVID onset (2020–2021)	54
40	Surge (2021–06 to 2023–12)	54
41	Disinflation (2023–2026)	54
42	Placebo — does the regressor ‘predict’ the past?	55
43	Household focus and each moment of the distribution	55
44	Same, restricted to quarters when expectations are moving	55
45	Does it have to be household focus? The three tone measures compared	56
46	Carroll’s second implication — does attention amplify it?	56
47	The randomisation test and why the nominal p-values must not be quoted	56
48	What actually survives, and what it means	56
49	Press household framing (reduced form) (s_news) — 59 quarters available	57
50	BoE household framing (reduced form) (s_boe) — 86 quarters available	57
51	Information shock (eps_cbi) — 86 quarters available	57
52	Stance shock (eps_mp) — 86 quarters available	58
53	Romer-Romer policy shock (policy_shock_std) — 42 quarters available	58
54	1. How much do they actually move together?	58
55	2. Who leads whom?	59

56	3. Lead-lag regressions (6 lags, own lags controlled, HC1)	59
57	Local projections on each component, run separately	60
58	Both components entered in the same regression	60
59	The clean experiment: pre–2015 minutes releases	60
60	Post–2015 (decision, minutes and MPR released together)	60
61	Window length	61
62	Placebo and pre-trend — windows before and around the event .	61
63	Placebo and pre-trend — controlling for the pre-window move directly	61
64	Null 5 — complexity and readability	62
65	Null 6 — term structure of inflation compensation	62
66	Null 7 — MPC dissent as a regressor	62
67	Null 8 — cumulative (three-month) framing surprise	62
68	Null 10 — monthly market tests: Dictionary hawkishness	62
69	Null 10 — monthly market tests: RoBERTa tone	63
70	Empirical size of the local projection under two estimators	63
71	Randomisation test on the attention interaction	63
72	Five-fold random CV	64
73	Blocked time-series CV	64
74	Penalty path	64
75	Seed sensitivity, five-fold random CV	64
76	Diagnostic: the cost of retaining a thinly observed candidate . . .	65

Abstract

Between 2005 and 2026 the mean UK household inflation expectation was 2.97% against realised inflation of 2.89%. Behind that mean, 22% of households expected inflation above 5%. Anchoring is therefore a property of the average household and not of the distribution behind it. Instruments for the textual analysis of central bank communication score it along the hawkish-dovish axis inherited from the dictionary literature, without identifying the audience a message addresses.

This dissertation measures audience. Four measures are built from Bank of England documents and 6,625 newspaper articles across five national titles, monthly from November 2005 to July 2026. Direction is measured by a word2vec-expanded stance dictionary and a fine-tuned transformer sentiment classifier, audience by the same transformer's agent classifier and a keyword ratio of household to market vocabulary. Each is projected onto the information already public at the date of publication, and the residual surprise is traced through lag-augmented local projections. Identification comes from a policy shock built on the Committee's own projections and a sign rotation on daily equity responses that separates stance from information.

The audience measures move household expectations, while the direction measures do not. Household-framed press coverage is followed by expectations 0.35 percentage points lower a year ahead, and Bank text by 0.26. Nothing moves before the communication and the effect builds only from the sixth month. The direction measures reach policy and markets instead.

The response is concentrated in the tails. Framing shrinks the share expecting 5% or more by 4.6 points and raises the share who say they do not know, while disagreement does not move at any horizon. The effect is largest when coverage is light, falling from 0.68 to 0.12 percentage points as attention rises. Communication reaches households through the audience it addresses and not the direction it signals, and it does so in the quiet periods, when inflation is not already the subject of daily coverage.

Acknowledgements

I would like to thank Professor Stephen Hansen for his guidance and feedback throughout this project, and in particular for the questions that shaped this study and took me far beyond its results.

Word count: 10,896

1 Introduction

Research on inflation has shifted towards the factors that form and anchor expectations, as the New Keynesian Phillips curve makes current inflation depend on expected future inflation: price and wage-setting decisions are forward-looking, so what households expect today shapes what is realised in future (Coibion, Gorodnichenko and Kamdar, 2018). If households expect higher inflation, they bargain for higher wages and accept higher prices, making expectations partly self-fulfilling and shaping consumption and saving decisions. Monetary policy therefore seeks not only to monitor inflation expectations, but also to influence how they are formed and anchored.

The conventional measures suggest that UK household inflation expectations are well anchored. Between 2005 and 2026 the mean one-year household expectation was 2.97%, against realised CPI inflation of 2.89% over the same period, a gap of 0.09 percentage points. The distribution behind that mean tells a different story. On average, 22% of households expected inflation above 5%, a share ranging from 7% to 48%, and only 48% considered the 2% target to be about right. Coibion and Gorodnichenko (2026) reach a similar conclusion for the United States, arguing that household expectations remain unanchored once the distribution rather than the mean is examined. This heterogeneity shifts attention from whether expectations are anchored on average to how central banks communicate with different audiences.

Central banks have responded by adopting modern communication approaches to reach more households and anchor expectations: plain summaries, social media announcements, and visual reports are all attributes of a modern central bank. Communication is now understood as a policy instrument in its own right instead of a supporting tool to the rate decision (Woodford, 2005; Blinder et al., 2008). Substantial literature establishes that central bank language carries information beyond the policy action itself (Gürkaynak, Sack and Swanson, 2005; Hansen, McMahon and Tong, 2019). The Bank of England has been among the most active, and whether it reaches households is an open empirical question. As the commitment to the inflation target is broadly credible, emphasising it has limited scope to move expectations, increasing the importance of framing and audience.

The measurement of central bank text has developed through three approaches. Dictionary methods count words from a specified list and remain the standard: Apel, Blix Grimaldi and Hull (2022) provide a widely used stance dictionary, following Loughran and McDonald (2011), who showed that sentiment dictionaries misclassify theme-specific text. Topic models collect topics from co-occurrence

rather than from a list, and Hansen, McMahon, and Tong (2019) apply Latent Dirichlet Allocation (LDA) to Inflation Reports. Transformer classifiers read each sentence in context and can be fine-tuned to a labelling task, as in Pfeifer and Marohl (2023) with CentralBankRoBERTa. Ash and Hansen (2023) show that these approaches frequently disagree when applied to the same corpus, in ways that affect regression inference. What they share is the object measured. Each scores communication on direction, how hawkish or dovish it is, and none asks whom the communication is about.

This dissertation measures audience rather than direction. The two corpora are Bank of England documents and 6,625 newspaper articles from five national titles. The sample begins in 2005, with the start of the monthly household expectations series, and falls entirely within the 2% CPI inflation-targeting regime adopted in December 2003. From these, four measures are constructed, two capturing direction and two capturing audience, checking the share of a document’s sentences about households, and the share of newspaper vocabulary drawn from household life. Each is orthogonalised against the macroeconomic information available at the time, including the professional forecasts, and entered into lag-augmented local projections. Two identification strategies are applied: a Romer and Romer (2004) policy shock constructed from the Committee’s own published projections, and a Jarociński and Karadi (2020) rotation using daily equity responses. The outcome is measured not only as the mean expectation but across the full binned distribution of household beliefs, which the Bank’s survey publishes and which allows the tails to be analysed separately.

Across the distribution, household framing shifts five of six moments together: the share expecting 5% or more falls by 4.6 points at eight quarters, and the share answering “don’t know” rises by 2.0 points. A stance shock has a near-zero effect on the mean, at -0.075 , but moves 1.4 percentage points out of the high-expectations tail and 1.7 percentage points into the low tail. These results connect with the literature on the tails of the inflation distribution (Adrian, Boyarchenko and Giannone, 2019; López-Salido and Loria, 2024), which finds the conditional mean uninformative about tail behaviour, and extend the same question to the distribution. The closest existing study, Tong and Cherry (2026), uses a stance dictionary and the mean one-year expectation, leaving the audience and the distribution outside its scope.

2 Literature and contribution

2.1 Communication and household expectations

Communication reaches markets and households through different channels and with different reliability. Market participants follow policy statements closely, and their responses appear in asset prices within minutes, which is what makes event studies with narrow windows possible (Gürkaynak, Sack and Swanson, 2005). Households neither seek this information nor receive it directly. Blinder et al. (2024) describe this mismatch as a sender without a receiver problem. Therefore, households receive central bank communication about inflation through intermediary channels such as media.

Carroll (2003) sees household expectations as absorbing the professional forecast through news coverage. In any period, a household either adopts the forecast reported in the press or keeps the view it already held, with the probability of updating fixed at λ . The response to news is delayed, it builds rather than arrives at once, and the gap between household and professional expectations narrows when coverage is heavy. This mechanism is tested in Section 6 and set against the results in Section 7.1.

According to Blinder et al. (2024), the dominance of intermediary channels means central banks must reach the public through the media, but that does not stop the Bank shaping the message. Media is not a neutral pipe. As Blinder et al. (2024) report, newspapers shape stories towards more extreme views, and reporting tends to overweight negative news. Coverage can therefore distort the message rather than transmit what was intended. This makes it useful to consider press framing and Bank framing separately, as done in Section 4.

Lamla and Maag (2012) find that German households' forecast disagreement depends on the heterogeneity of story content and reporting intensity, particularly for news on rising inflation, while professional forecasters' disagreement does not depend on media coverage at all, similar to the findings in Section 6.6. Larsen, Thorsrud and Zhulanova (2021) show that the topics the media report on predict both inflation and household inflation expectations, and that the time-series behaviour of those topics accounts for information rigidity that varies over time. The result concerns the distribution rather than the mean, which Section 6.6 also explores.

Tong and Cherry (2026) bring this to UK data. They measure each announcement against the press coverage that preceded it to obtain a narrative surprise, separate it into a stance component and an information component (Jarociński and Karadi,

2020) and find that the two components move household inflation expectations in opposite directions, with the information component strongest when public attention to inflation is high, and that shocks identified from asset prices alone do not move households. These results are tested in Section 6.5.

What these measures share is the quantity they score. Larsen, Thorsrud and Zhulanova (2021) identify the earlier literature as counting inflation terms to capture the intensity of coverage, and replace that count with the composition of topics. Stance dictionaries replace it with direction. Neither asks whom the communication is about, and it is what the dissertation measures.

2.2 Measuring text

Text has to be turned into a quantitative measure to enter a regression or to be documented and compared against other variables. The first step is common to all: documents are split into tokens, tokens are reduced to stems, and what remains is a vector of counts. Dictionary methods fix the measurement rule in advance by counting word occurrences relative to document length. The literature shows how much the resulting measure depends on word choice. Loughran and McDonald (2011) show that general-purpose sentiment lists misclassify financial text, as the same word carries different meanings in different contexts. They find that almost three-quarters of the words counted as negative by a general-purpose dictionary are not negative in a financial setting. Apel and Blix Grimaldi (2012) extend this approach to central banking with a hawkish/dovish stance dictionary.

The word-embedding approach expands the word list from the corpus rather than specifying it entirely in advance. Ash and Hansen (2023) describe the procedure Section 4.1 adapts: the researcher selects seed words for a concept, uses word embeddings to identify the terms nearest them, and filters the resulting list. As they note, this approach leaves polysemy (the problem of a word having several meanings) unresolved. Each word has a single vector combining its meanings, so a term selected for one meaning contributes to the score regardless of its meaning in context. Section 4.4 documents the consequences over a sample of two decades.

Baker, Bloom and Davis (2016) count articles containing a triple of terms about the economy, policy and uncertainty, and treat frequency as the index, scoring no word as good or bad, so that the measure captures the focus of the document rather than its sentiment. This is the construction used for the press framing measure in Section 4.3. They validate it against a human audit of twelve thousand articles, and report that indices built separately from right-leaning and left-leaning titles move closely together, which they read as evidence that political orientation does

not distort the index. Kalamara et al. (2022), forecasting UK output, inflation and unemployment from newspaper text, find that simple word counts perform surprisingly well against more elaborate alternatives.

When the concept is not named in advance, the literature turns to topic models. Hansen, McMahon and Tong (2019) apply LDA to Bank of England Inflation Reports to represent each Report as a distribution over topics, and combine those narrative signals with the numeric ones, the inflation and GDP forecasts, testing whether the narrative carries news beyond the numbers by comparing the selection against random reorderings of the yield residuals. Sections 5.3 and 6.4 borrow that logic, comparing an estimate against a distribution generated under the null.

The most recent method reads each sentence in the context of the words around it. Pfeifer and Marohl (2023) fine-tune RoBERTa on central bank language for two tasks, one sorting sentences by the economic agent they concern and one by sentiment, reaching weighted precision. Both are used in Section 4.2. Their labelled sentences are drawn entirely from Federal Reserve speeches, and the sentiment labels are positive and negative rather than hawkish and dovish, delivering different results.

Ash and Hansen (2023) apply ten algorithms to the same corpus and find that they disagree about it, in ways that carry through to the coefficients and standard errors of whatever regression they feed, and conclude that economics research has no agreed benchmark against which the choice could be validated. What remains available is external validation, either running two independent constructions of the same concept over the same text and asking whether they agree, or testing whether a measure predicts something it was not built to predict. Sections 4.4, 6.2 and 6.3 do both.

2.3 Identification

Communication and policy are endogenous, as both arise in response to economic conditions. What follows a statement is therefore not caused by the statement alone, and the identification is to isolate the part of it that the economy did not already imply. Kuttner (2001) splits target rate changes into anticipated and unanticipated components using federal funds futures. He finds the response of market rates to anticipated changes small and the response to unanticipated changes large and highly significant. Gürkaynak, Sack and Swanson (2005) show that a single factor, the change in the target rate, does not capture the response of asset prices, and a second is required, which they interpret as the future path of policy (forward guidance) and associate with statements rather than with

decisions.

The orthogonalisation in Section 5.1 is the substitute for cases where narrow windows are not available. Each text measure is projected onto the macroeconomic information available at the time, retaining the residual, which removes only what the chosen controls capture. Romer and Romer (2004) replace the researcher’s control set with the policymaker’s, regressing changes in the intended federal funds rate on the Federal Reserve’s own forecasts of output and inflation so that the residual is policy the Committee’s own view does not explain. Section 5.4 applies the same logic to the Monetary Policy Committee’s published projections.

A surprise identified in either way contains a mixture of shocks with different effects on expectations. Jarociński and Karadi (2020) separate the stance shock and the information shock by the co-movement of interest rates and equity prices in the announcement window. A tightening raises rates and lowers equity prices, since the discount rate rises and expected payoffs fall, so opposite signs identify a policy shock and matching signs an information shock. Section 6.5 applies the rotation to a surprise that did not give significant results before it.

These methods identify what the shock is rather than how its effect is estimated. Jordà (2005) estimates impulse responses by a separate regression at each horizon rather than by iterating a fitted VAR forward, so that a misspecification at a single horizon does not compound. Montiel Olea and Plagborg-Møller (2021) establish the inference: lag-augmented local projections are valid uniformly across stationary and non-stationary data and across a wide range of horizons. Lag augmentation removes the need to correct standard errors for serial correlation. Sections 5.2 and 5.3 follow both.

3 Data

The collected data fall into three groups: published statistical and market series, inflation expectations survey series, and two text corpora assembled for this project, aligned to a monthly panel in which the household expectation is the dependent variable and the text measures are the regressors. The monthly sample runs from November 2005, when the household expectations series begins, to July 2026. The quarterly survey covers 94 quarters over the same period, and the newspaper corpus covers the 172 months for which it was collected.

3.1 Household inflation expectations

The Citi/YouGov monthly household inflation expectations survey reports a point estimate of twelve-month-ahead expectations. It publishes no breakdown of responses, so dispersion is not observable. It is nonetheless the only monthly UK series available and the one the Bank monitors, and monthly frequency provides eighteen horizons for the local projections, allowing for a better-resolved impulse response. Every monthly specification in this dissertation uses Citi/YouGov as the dependent variable.

The BoE/Ipsos quarterly Inflation Attitudes Survey covers 94 quarters of approximately 2,000 respondents each. Respondents select a band rather than a number, and the share choosing each band is published, so the full distribution is recoverable. From it the following are obtained: disagreement (the standard deviation across bins), the shares expecting above 5% and below 2%, target credibility (the share replying that the 2% target is about right), the ‘don’t know’ share, and the five-year question used for long-horizon de-anchoring.

The ‘don’t know’ share is an attention measure taken from inside the survey instrument itself, so it is independent of the Google Trends and Nexis measures used elsewhere, and it provides a check on the household updating rate in Carroll (2003) that does not rely on media data.

The two surveys are distinct instruments tracked over 2005–2026, and using both is what makes the comparison between the mean and the distribution in Section 6 possible. Because their methodologies and samples are not the same, differences between them cannot be attributed to moments alone.

3.2 Bank of England text

The Bank of England text corpus consists of 312 sets of Monetary Policy Committee minutes, which record the discussion and the vote split, and 95 Monetary

Policy Reports, which contain the Committee’s forecast and its assessment of the economy. The Bank was granted independence in May 1997, and the first MPC meeting took place the following month, so the minutes begin in June 1997. The Reports in the corpus begin in February 2003. Documents were downloaded as PDFs from the Bank’s website, converted to text, and stripped of headers, footers and standard annexes, since the measures constructed in Section 4 are computed as shares of text and unnecessary material would distort the measures. The dictionary in Section 4.1 is built using a word2vec model trained on approximately 8.9 million tokens from the corpus. As the dependent variable (Citi/YouGov Survey) begins in 2005, 308 documents fall within the estimation window.

Before August 2015, the Bank published meeting minutes two weeks after announcing the rate decision. From August 2015 the ‘Super Thursday’ arrangement ensured that the decision, the minutes and, in Report months, the Report itself have been released simultaneously, so that the Committee’s reasoning reaches the public at the same moment as its decision. The earlier schedule is useful for identification. For pre-2015 minutes publication dates, there is no policy announcement, since the decision is made earlier and is already priced by the markets. Anything that moves in a window around that date is therefore related to the text itself, which is an unusually clean experiment in this literature. After 2015 no such separation is possible: a move on a Super Thursday could reflect the decision, the vote split, the forecast or the language.

The corpus is used in three ways. The text measures of Section 4 are computed on every document in the estimation window, the rotation in Section 6.5 treats all 308 publication dates as events, and the event study in Section 6.3 uses pre-2015 minutes only. The effect estimated there is absent after 2015, which displays the loss of identification due to the change of the announcement schedules rather than a change in the effect of communication.

3.3 Newspaper text

The newspaper corpus draws on five British media sources: the Financial Times, the Guardian, the Times, the Telegraph and the Daily Mail. They were selected according to the political range of the press, but also to include the papers most read by the households whose expectations this dissertation measures. A corpus of broadsheets alone would consist largely of market-facing coverage and would bias the framing measure against the household vocabulary it is designed to detect.

Articles were retrieved monthly from the Nexis news database using inflation-related search terms, selecting the highest-ranked results by the platform’s rel-

evance score, which measures the percentage of coverage of the topic. 11,192 articles were retrieved and reduced to 8,702 after removing near-duplicates and to 6,625 after applying a uniform sampling rule.

The sampling rule addresses a second issue: monthly sampling rates were correlated with the framing measure itself. Before the rule was introduced, the share of available articles retained ranged from 4.2% to 14.2%, linking the construction of the regressor to the value it was intended to measure. The rule therefore retains a share of 5% of distinct articles with a floor of 25 per month, and drops duplicates before sampling. The floor is set at 25 rather than higher because a larger floor binds in almost every month and returns the rule to a fixed count, while at 25 it binds in roughly a quarter of months and the rule stays proportional across the rest. A rate of 5% achieves the same band of selectivity as 6% or 8% at a lower collection cost.

Nexis licensing prohibits redistribution, so the replication materials contain the code and the derived monthly series but not the article texts. Both corpora pass through the same preparation before any measure is computed: Bank documents are stripped of boilerplate, articles are parsed into title, date, source and body, and the text of each is lower-cased and split into sentences and tokens. Appendix A2.3 sets out the steps. The framing measure built from this corpus is described in Section 4.3.

3.4 Macroeconomic, market and policy series

Macroeconomic and market series were obtained from the Office for National Statistics (CPI, GDP and unemployment), the Bank of England (Bank Rate and the implied inflation curve), LSEG Refinitiv (commodity prices, swaps and breakevens), GfK (consumer confidence) and Reuters (forecaster polls). Series enter as published. CPI and its components are twelve-month rates and Bank Rate is the end-of-month level, so no seasonal adjustment is applied. Daily series are collapsed to a monthly frequency by two rules: swaps and breakevens take the last quote of the month, since a month-end value is a clean read of what the market expected on that date, while gas prices take the monthly mean, since they enter as the cost faced over the month. Nothing is interpolated or carried forward, and each regression is estimated on the months for which every variable it uses is observed.

CPI, Bank Rate, the five-year five-year inflation forward, NBP gas prices and the Reuters poll of professional forecasters form the first-stage information set. Monthly changes in CPI, Bank Rate and GfK consumer confidence are the controls

in the local projections. The implied inflation curve at maturities from 2.5 to 20 years, five-year breakevens and the Bank Rate serve as outcome variables in Section 6. Google Trends searches for inflation-related terms serve as an alternative attention measure, and a cross-check on the Nexis counts.

Two further series enter the identification. Projections for CPI and GDP were taken from the Monetary Policy Reports published between August 2013 and February 2026, as the Reports presented projections only as fan charts before that date, with no numeric table. Daily FTSE 100 closes over 5,463 trading days are used to sign the equity response in the rotation of Section 5.4. As with the Nexis material, the LSEG series are licensed and cannot be redistributed. Appendix 2 provides a full list of variables, definitions, and sources.

4 Measurement

4.1 A dictionary measure of stance

The existing literature measures the direction of communication using dictionaries of hawkish and dovish language. Building one for this corpus means expanding a small seed list computationally, since writing a long list by hand would impose the answer in advance. The initial set contains seven hawkish and seven dovish seeds, chosen on semantic grounds and corpus frequency before any regression was estimated. The model was trained on the Bank and newspaper corpora combined, since the design requires one instrument that scores both comparably. Trained separately, the hawkish vocabularies of the two corpora overlap by only 10-12% among the top sixty words. The Bank and the press use different language to convey the same stance, so two separately-built dictionaries would have been near-disjoint instruments and no comparison between them would have been meaningful.

Word2vec assigns each word a vector learned from the contexts in which it appears, so that words used in similar contexts acquire similar vectors. Candidates are ranked not by similarity to the hawkish seeds but by their projection onto the axis running from the mean dovish vector to the mean hawkish vector, since antonyms occur in almost identical contexts. The full dictionary classification is reported in Appendix 1. Only hawkish stems enter the score, while the dovish classifications were used to keep those words out of the hawkish list.

The score is the number of hawkish stem tokens as a percentage of tokens in inflation-related sentences. The score is defined as

$$\text{hawk}_d = 100 \times \frac{\text{hawkish stem tokens in inflation-related sentences of document } d}{\text{all tokens in those sentences}}. \quad (1)$$

Restricting the count to those sentences is necessary because the dictionary cannot read context. Unrestricted, the measure scored 2020 at 0.53 against CPI of 0.8%, higher than 2011 when CPI was 4.5%, because some words are in the dictionary as signals of alarm about inflation but in 2020 describe the pandemic. The unrestricted measure reveals the intensity of concern language rather than the hawkishness of monetary policy. Confining both numerator and denominator to inflation-related sentences also keeps stance separate from attention, which is measured independently in Section 3.3.

4.2 Classifier measures

CentralBankRoBERTa (Pfeifer and Marohl, 2023) provides an alternative to the dictionary measure in Section 4.1. It applies the general-purpose RoBERTa model, fine-tuned on sentences hand-labelled from Federal Reserve speeches, with no pretraining on central bank text. Two supervised classifiers are fitted on it: a sentiment classifier hand-labelled on 6,683 sentences and an economic agent classifier on 6,205, each then expanded by self-training, so most of the training data carries machine rather than human labels. The published weights were applied unchanged, and no parameter was estimated on the corpora used here, which removes any possibility that the instrument was tuned towards a particular result.

The two classifiers answer different questions. The sentiment classifier assigns each sentence a positive or negative label. The agent classifier assigns each sentence to one of five categories according to whom it concerns: households, firms, the financial sector, the government, or the central bank itself. The household focus measure used throughout is therefore the share of a document allocated to households out of five competing categories. A dictionary registers a sentence only when a listed word appears in it, whereas the classifier reads the sentence in context, so that a passage about the cost of the weekly grocery shop is recognised as concerning households whether or not it contains the word “household”. No keyword list can capture that, because the vocabulary a newspaper uses to describe domestic economic life is open-ended. This classification was aggregated to a document and monthly shares, covering 181,861 sentences across 407 documents. Averaged across documents, the Bank devotes 36.7% of its sentences to firms, 22.5% to itself, 20.6% to households, 16.5% to the financial sector and 3.7% to government. Households are therefore not the Bank’s principal subject, and it writes about firms nearly twice as often.

4.3 Press household framing

The press measure identifies which problem inflation coverage emphasises. It is the count of household-frame words as a share of household- and market-frame words combined. Formally

$$\text{focus}_m = \frac{\text{household-frame terms in month } m}{\text{household-frame terms} + \text{market-frame terms in month } m}. \quad (2)$$

The measure distinguishes between the household frame that concerns domestic economic life, such as cost of living, grocery, wage, earnings, mortgage, rent, energy bill and food price, and the market frame, which captures the language of

finance: gilts, yields, equities, the FTSE, margins and balance sheets. The ratio isolates framing rather than attention, since a raw count would rise whenever inflation coverage grew, and the volume of coverage is measured separately by the Nexis counts described in Section 3.3. Both lists were fixed on semantics before estimation, as with the dictionary in Section 4.1, and are reported in full in Appendix 1.

The agent classifier in Section 4.2 could be applied to the same articles, but it was fine-tuned on central bank speeches, and newspaper prose differs. A keyword measure is also auditable and follows the construction used by Baker, Bloom and Davis (2016). A classifier version is reported as a robustness check in Section 6.2.

Across 172 months, the measure averages 0.527 with a standard deviation of 0.139, ranging from 0.169 to 0.843. It is built from 57,639 household-frame and 52,156 market-frame hits across 6.3 million words in 6,625 articles. The two vocabularies are close to balanced on average, and the range is wide enough that the measure never settles near either extreme.

The measure moves with episodes that can be named. Annual means are high in 2008 (0.648), 2011 (0.640) and 2014 (0.675), each a period of tight household incomes, and low through the pandemic, at 0.364 in 2020 and 0.324 in 2021, when inflation was discussed through supply chains and financial markets. The measure then rises to 0.597 in 2022, the year the cost-of-living crisis entered everyday language.

4.4 Validation

Constructed text measures require independent validation. Three of the four measures satisfy this requirement, while the dictionary measure does not. The first tests the convergence: two independent constructions of the same concept should agree. The press framing measure was rebuilt using the CentralBankRoBERTa agent classifier on the same 6,625 articles, and the two series correlate 0.794 in levels and 0.724 in changes. The constructions share no components, as one applies a specified vocabulary and the other a published classifier that requires none, using pretrained weights, so the agreement cannot follow from common construction and indicates that both capture variation present in the text. Section 4.3 provides a second check on the same measure, in annual means that track episodes identifiable independently of the data.

The second test asks the opposite question, whether measures that should be distinct in fact are. The Bank and press framing measures correlate -0.007 in levels, and their orthogonalised surprises correlate -0.003 . The two measures

therefore track different variation, and results estimated on the two corpora provide independent evidence.

The third test is whether a measure predicts what it claims to measure, and here the dictionary fails. Its correlation with the next policy rate change is $+0.027$, carrying no information about what the Committee did. Its mean rises by a factor of 5.6 across the sample, and it peaks during the 2024-26 cutting cycle, which is the wrong direction for a measure of hawkishness. Dictionary methods assume a stable path from vocabulary to stance, and over a two-decade sample that assumption fails as lexical change in institutional language becomes equivalent to variation in the stance itself. Ash and Hansen (2023) show that alternative text algorithms applied to the same corpus frequently disagree in ways that affect regression inference, and that fine-tuned pretrained models generally dominate bag-of-words approaches, including dictionary methods.

A fourth test asks the same question of the direction dimension. Because the dictionary was trained on both corpora together, one instrument scores Bank text and newspaper text, so the two can be compared directly. In levels, they correlate $+0.052$, and no lead or lag out to six months in either direction raises the correlation above 0.124. In monthly changes, the correlation is $+0.270$, significant at 5%. The press moves with the Bank's stance at the margin without relaying it, which is what Blinder et al. (2024) describe when they report that coverage distorts the message. Press framing is therefore an independent channel and not a restatement of the Bank channel, which is what the comparison in Section 6.1 requires.

5 Empirical strategy

5.1 Constructing the surprise

This section turns the four text measures constructed in Section 4 into regressors. Raw measures cannot be used directly in the regression, as framing responds to the economy. For example, during an energy price shock, the press writes about rising bills, so the household framing measure would partly be a proxy for energy prices, and any correlation with expectations might reflect those prices and not how inflation was presented.

Each measure is therefore regressed on the macroeconomic information available at the time: CPI, Bank Rate, the five-year five-year inflation forward, NBP gas prices and the Reuters poll of professional forecasters, and the residual is divided by its own standard deviation:

$$m_t = \alpha + \gamma' \mathbf{I}_t + u_t, \quad x_t = \frac{u_t}{\hat{\sigma}_u}, \quad (3)$$

where m_t is the raw text measure, $\mathbf{I}_t$ the public information set listed above, and x_t the standardised surprise.

A value of $x_t = 1$ therefore denotes coverage one standard deviation more household-focused than the macroeconomic data and the professional consensus would have predicted. Standardising makes the coefficients comparable across text measures, which are constructed on different raw scales. The Reuters poll matters as in the literature, Carroll (2003) treats the professional forecast as the signal that reaches households through the media, so conditioning on it ensures the surprise reflects how inflation was presented rather than what forecasters were saying. This follows the practice of Romer and Romer (2004), who use policy regressions with the Federal Reserve’s own Greenbook forecasts. The same logic is applied in Section 5.4, where the Bank’s published projections replace the professional consensus.

The share of variation surviving this projection is 77.6% for the dictionary, 83.4% for both classifier measures and 89.7% for press framing. This establishes only that the measures are not mechanical functions of observable macro data. Section 6.3 tests the claim that the measures contain information by using them to forecast outcomes.

5.2 Local projections

The variable of interest is the path of household inflation expectations following a framing surprise, rather than a contemporaneous association between the two series. The distinction matters because the timing is itself a test. Carroll (2003) predicts a delayed and building response, since households revise their beliefs only as they face news coverage, so an effect appearing on impact could be caused by simultaneity. For this reason, the analysis uses local projections (Jordà, 2005), which estimate a separate regression at each horizon rather than imposing a dynamic model. A vector autoregression estimates a single lag structure, iterating and forecasting it forward, so misidentification at one horizon compounds across horizons, whereas local projections estimate each horizon independently with a separate regression.

For each horizon h :

$$y_{t+h} - y_{t-1} = \alpha_h + \beta_h x_t + \phi_h x_{t-1} + \rho_{1h} y_{t-1} + \rho_{2h} y_{t-2} + \delta'_h \Delta \mathbf{Z}_t + \varepsilon_{t+h}, \quad (4)$$

where y_t is the Citi/YouGov one-year household inflation expectation, x_t the standardised surprise defined in Section 5.1, and $\Delta \mathbf{Z}_t$ contains monthly changes in CPI, Bank Rate and GfK consumer confidence. Each horizon is estimated as a separate regression, so all coefficients are horizon-specific; only β_h is reported. The dependent variable is the cumulative change from the month before the shock to h months after, so plotting $\{\hat{\beta}_0, \dots, \hat{\beta}_H\}$ against h traces the impulse response. A flat, non-zero path at longer horizons therefore marks a persistent shift, not an effect that keeps growing.

Horizons run from $h = 0$ to 18 months, or from 0 to 8 quarters, where the Ipsos distribution is the dependent variable. The regression includes two lags of the dependent variable and one lag of the regressor. The second lag implements the lag augmentation of Montiel Olea and Plagborg-Møller (2021) for the reasons set out in Section 5.3. The framework also accommodates the state-dependent specification of Section 6.4 and the distributional dependent variables of Section 6.6.

5.3 Inference

Local projections create a problem for inference. At horizon h the dependent variable is the cumulative change from $t - 1$ to $t + h$, so the observations overlap. The observation for month t goes from $t - 1$ to $t + h$, and the one for month $t + 1$ spans t to $t + h + 1$, sharing h months of the same data. The residuals

are therefore serially correlated, and standard errors that assume independent residuals will be biased. The conventional solution is a HAC estimator such as Newey-West, which is what the literature typically uses.

To evaluate the performance in a sample of this size, the approach follows Bertrand, Duflo and Mullainathan (2004), who assess the size of an estimator by generating placebo interventions at random and recording how often the null is rejected. The exercise applies a test with a nominal size of 5% to 1,000 artificial regressors, each generated as an AR(1) process matched to the persistence of the framing surprise and unrelated to household expectations. Matching the persistence is important, since persistence is what degrades these estimators in finite samples. Newey-West rejected the null in 12.0% of draws at a twelve-month horizon and 15.3% at eighteen months, against a nominal 5%. The over-rejection widens with the horizon, as the lag window grows while the number of observations falls.

Table 1: Empirical size of the local projection under two estimators

Horizon	Newey–West, no lag augmentation	Lag-augmented, HC1
$h = 6$	8.3%	7.4%
$h = 9$	10.5%	6.8%
$h = 12$	12.0%	6.2%
$h = 18$	15.3%	7.5%

Note. Share of 1,000 AR(1) placebo regressors, matched to the persistence of the real regressor ($\rho = 0.461$) and unrelated to household expectations by construction, for which the null is rejected at the 5% level. A correctly sized test rejects 5% of the time.

The specification therefore follows Montiel Olea and Plagborg-Møller (2021). Two lags of the dependent variable and one of the regressor enter the right-hand side, and inference uses heteroskedasticity-robust HC1 errors with no serial-correlation adjustment. Their result is that this procedure is uniformly valid, and its coverage does not depend on the persistence of the data, which is unknown. Repeating the size experiment under this specification gives rejection rates of 6.2% at twelve months and 7.5% at eighteen, close to each other.

5.4 Identification

To identify the effect of communication, three strategies are used, each addressing a limitation of the previous. The first, set out in Section 5.1, projects each text measure on the macroeconomic information available at the time and retains the residual. Its limitation is that it can remove only what the chosen macroeconomic

controls capture, which can be argued, and the correct control set can vary across time periods.

Romer and Romer (2004) address this by replacing the control variables set with the policymaker’s forecasts. Policy responds to what the central bank expects, so an observed rate change contains a systematic component reflecting the Committee’s reading of the outlook.

$$\Delta R_t = \alpha + \beta R_{t-1} + \sum_h \gamma_h F_t[\pi_{t+h}] + \sum_h \delta_h F_t[g_{t+h}] + \sum_h \lambda_h \Delta F_t[\pi_{t+h}] + \sum_h \theta_h \Delta F_t[g_{t+h}] + \varepsilon_t, \quad (5)$$

where $F_t[\cdot]$ denotes the projection published in the Report at t and $\Delta F_t[\cdot]$ its revision since the previous Report.

The change in Bank Rate is regressed on the Monetary Policy Report’s projections for CPI inflation and GDP growth, on revisions to those projections since the previous Report, and on the level of Bank Rate before the meeting, which enters because the systematic response depends on the current rate. The residual is the part of the decision that the Committee’s own published view does not explain. Romer and Romer use the Federal Reserve’s internal Greenbook forecasts, unavailable to the public at the time and therefore closer to what the policymaker actually knew than any published projection. The Monetary Policy Report is the published document, so it is narrower than the Committee’s full information set, and anything known but not disclosed remains in the residual. A second difficulty affects the text surprises themselves. A hawkish statement can mean two things: that policy will be tighter than expected (Odyssean forward guidance), which should lower inflation expectations or that the Committee believes the economy is stronger than the public had assumed (Delphic forward guidance), which should raise them. Both read as hawkish, so a regression on their sum estimates a weighted average of two coefficients of opposite sign, which offset one another, producing an estimate closer to zero that represents neither response clearly.

Jarociński and Karadi (2020) separate the two through their opposing effects on equity prices: equities fall following a surprise tightening but rise when the surprise reveals a stronger economy. Over the window $[d - 1, d + 1]$ around each of the publication dates, a text surprise accompanied by an equity move of the opposite sign is classified as a stance shock, and one accompanied by a move of the same sign as an information shock. The identifying assumption is that these are the only two disturbances moving both series within the window, which is

standard in this literature. Both components then enter the same local projection, with the predicted signs, $\beta^{MP} < 0$ and $\beta^{CBI} > 0$, recorded before estimation.

6 Results

6.1 The audience channel

Whether communication about households reaches households is the first claim to establish. Table 2 reports the response of the Citi/YouGov one-year household inflation expectation to a one-standard-deviation household-framing surprise, estimated by lag-augmented local projections with HC1 standard errors. A one-standard-deviation increase in the orthogonalised household-framing component of press coverage is followed by expectations 0.346 pp lower twelve months later. The same specification applied to Bank of England text, using the Central-BankRoBERTa agent classifier, gives -0.255 pp, although the two framing series correlate -0.007 in levels. Over 2005–2026 the Citi/YouGov series averages 2.97%, with a standard deviation of 1.11 pp and a range of 0.8% to 6.3%. The standard deviation of its own twelve-month change is 1.13 pp. The press coefficient is therefore under a third of a standard deviation of both the level of the series and its typical annual movement.

Table 2: Household framing and one-year household inflation expectations

Regressor	$h = 6$	$h = 9$	$h = 12$	$h = 18$	n
Press household framing	-0.128^*	-0.284^{***}	-0.346^{***}	-0.378^{***}	143
Bank household framing	-0.213^{***}	-0.237^{***}	-0.255^{***}	-0.147^*	156

Leads, press household framing:

three months before	$+0.035$	$(p = 0.235)$
six months before	$+0.019$	$(p = 0.712)$

Note. Dependent variable: the Citi/YouGov one-year household inflation expectation, in percentage points. Monthly, 2005–2026. Lag-augmented local projections, equation (4), with HC1 standard errors. Coefficients are the response to a one-standard-deviation surprise. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

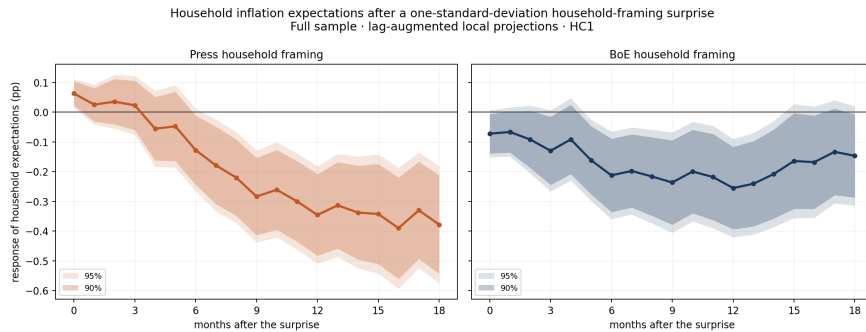


Figure 1: Response of the one-year household inflation expectation to a one-standard-deviation household-framing surprise. HC1 errors, 90% and 95% bands.

The results display that the response is delayed. On impact, the press coefficient is small and positive. It turns negative in the sixth month and is significant by the ninth. Figure 1 shows the full path. Delay is what the epidemiological model in Carroll (2003) predicts. An effect appearing on impact would in fact be harder to interpret, since coverage and expectations would then be moving together rather than one following the other.

The effect flattens, changing little between the twelfth and eighteenth months, which is the profile of a one-off level shift, not a regression fitting a common trend. The two measures differ in timing, revealing that Bank texts act sooner and fade by the eighteenth month. In contrast, press framing acts later and persists, consistent with the Bank’s own language being reported immediately and the press effect accumulating through repetition. Coincident measurement error cannot generate agreement of this kind. However, it does not rule out a common macroeconomic driver that survives the orthogonalisation and moves both the text and expectations together. The subsections that follow address that possibility in turn.

The surprise construction removes variation predictable from the public information set. The threat that remains is that framing responds to where expectations were already moving in the direction that the information set does not capture. Estimating the same projection at negative horizons places the dependent variable in a window that closed before the surprise occurred, so a non-zero coefficient would indicate a regressor tracking a pre-existing path. Here it does not, showing $+0.035$ ($p = 0.235$) three months before, and $+0.019$ ($p = 0.712$) at six. The sign is negative, meaning coverage framed around households is followed by lower expectations, not higher, even though such coverage largely concerns the cost of living. Whether that reflects reassurance or a narrowing of attention is not seen from a conditional mean. Section 6.6 returns to it with the full distribution of household beliefs from the survey.

6.2 Is the result an artefact of the word list?

A measure built from a word list is only as credible as the person who wrote it, so additional tests are necessary. The word lists were fixed on meaning before estimation, and the sampling rule of Section 3.3 removes the remaining source of variation in what the corpus contains. The impulse responses of household expectations to the two measures are very similar from the sixth month onwards. At twelve months, the keyword ratio gives -0.346 and the classifier -0.347 . The two series correlate at 0.794 in levels, and 0.793 in orthogonalised surprises. This rules out the word list as the source of the result, because the classifier has none.

The measures diverge only on impact, where the keyword ratio is positive and significant, while the classifier is close to zero, so the initial positive coefficient reported in Section 6.1 is due to the list rather than household framing.

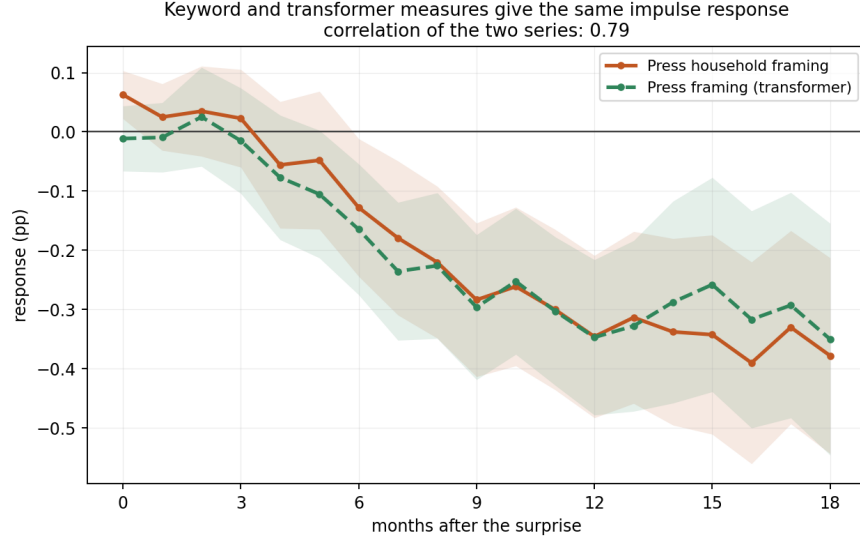


Figure 2: The same specification with the press framing measure built two ways: a hand-specified keyword ratio and the CentralBankRoBERTa agent classifier applied to the same 6,625 articles.

Validation against an independent instrument doesn't rule out the scenario in which both measures respond to a common feature of the text. A penalised regression provides another check for the regressors. Larsen, Thorsrud and Zhulanova (2021) use LASSO to link news content to inflation expectations and test whether the relationship remains informative after accounting for roughly 130 hard economic indicators. LASSO minimises squared prediction error subject to a penalty on the sum of the absolute coefficients, and because that penalty is not smooth at zero, it drives weak coefficients exactly to zero, so estimation and variable selection happen in a single step. Run here over 45 candidates observed for at least 80% of the panel, five-fold cross-validation retains twelve, among them press household framing and Bank household framing. Under blocked cross-validation, which respects the time ordering and is the harder test on a serially correlated panel, the penalty rises and three survive: food CPI, Brent, and press household framing. The measure is retained at every penalty at which more than two candidates remain. Without post-LASSO inference, this exercise only supports selection without performing inference.

6.3 Does communication carry information?

The first stage R^2 establishes only that the surprise is not a function of the controls, which the design requires but which is not evidence for it. Whether communication carries information is a separate question, and it can only be answered if a surprise predicts outcomes that have not yet occurred.

Table 3: Text surprises as predictors of macroeconomic and market outcomes

Outcome	Regressor	$h = 6$	$h = 12$	$h = 18$
CPI inflation	Bank household framing	−0.252**	−0.450***	−0.257
CPI inflation	Press household framing	−0.142	−0.490**	−0.765***
Bank Rate	RoBERTa sentiment	+0.272***	+0.435***	+0.504***
5y5y inflation forward	Press household framing	−0.062**	−0.089***	−0.035
5-year breakeven	Dictionary hawkishness	+0.068	+0.202***	+0.111*
5-year breakeven	Press household framing	−0.087	−0.249***	−0.237***

Note. Lag-augmented local projections with HC1 standard errors. Each row is a separate outcome variable and each cell a separate regression. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The direction measures predict policy even though they do not move households. RoBERTa sentiment forecasts Bank Rate at every horizon, by +0.272 at six months, +0.435 at twelve and +0.504 at eighteen, each significant at 1%. Dictionary hawkishness leaves the policy rate unmoved but raises five-year breakevens by +0.202 at twelve months. Both measures therefore carry information, and that information reaches markets and anticipates policy. Neither moves households, which is the null reported in Section 6.1.

Household framing predicts realised CPI inflation. Bank framing: −0.252 at six months, −0.358 at nine, −0.450 at twelve. Press framing: −0.490 at twelve, −0.765 at eighteen. The same surprise is followed by expectations 0.35 pp lower at twelve months, and by realised inflation lower at the same horizon. Expectation and outcome move in the same direction at the same horizon, which pure framing would not do.

Table 4: Press household framing and the implied inflation curve, by maturity

Maturity	$h = 6$	$h = 9$	$h = 12$	$h = 18$
3-year	-0.191**	-0.295***	-0.261***	-0.181**
5-year	-0.151***	-0.233***	-0.193***	-0.141**
10-year	-0.109***	-0.165***	-0.142***	-0.090**
15-year	-0.070**	-0.111***	-0.099***	-0.060*
20-year	-0.035	-0.061**	-0.049**	-0.020

Note. Dependent variable: the Bank of England implied inflation curve at each maturity, in percentage points. $n = 137$ to 143. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Hansen, McMahon and Tong (2019), using Inflation Report publications, find the opposite. The signals they extract move long-run rates without affecting short-run rates, and operate primarily through the term premium, because communication changes the perceived variance of future conditions. The two measures therefore recover different objects from the same text, and the classification target is what separates a measure of audience from a measure of uncertainty.

The monthly evidence has a daily counterpart. Before August 2015, the Bank published the minutes of a meeting roughly two weeks after announcing the decision, so a publication date carried text and nothing else, as mentioned in Section 3.2. On those dates, the RoBERTa measure moves the implied inflation curve by +2.28 basis points at two and a half years, +1.99 at three and +1.38 at five, while the dictionary measure is null at every maturity. The same regression estimated over a window from four to two days before publication gives +1.437 basis points, larger than the +1.195 measured over the event window itself. Something is already moving the curve before the text is released, so the publication-day response cannot be attributed to it.

Table 5: Publication-day response of the implied inflation curve, pre-2015 minutes

Measure	2.5-year	3-year	5-year
CentralBankRoBERTa sentiment	+2.28**	+1.99**	+1.38**
Dictionary hawkishness	−0.52	−0.65	−0.43
<i>Placebo windows, CentralBankRoBERTa sentiment, 3-year:</i>			
[−6, −4] before publication	+0.322	($p = 0.473$)	
[−4, −2] before publication	+1.437	($p = 0.031$)	
[−1, 0] event window	+1.195	($p = 0.077$)	

Note. Change in the Bank of England implied inflation curve, in basis points, over a two-day window around publication. Pre-2015 minutes only, when the rate decision preceded publication by approximately two weeks. HC1 standard errors, $n = 91$ to 122. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

6.4 When does the channel operate?

According to Carroll (2003), the inflation update rate of households should rise with media coverage, because more coverage brings the professional forecast to more households. The prediction is tested by scaling the framing surprise with standardised press attention to inflation and entering both terms, along with their interaction. The interaction is positive, and that is the opposite of what Carroll predicts. Because the main effect is negative, a positive interaction means the effect gets weaker as attention rises. Evaluated at twelve months, a one-standard-deviation framing surprise moves expectations by 0.68 pp when attention is one standard deviation below its mean, by 0.40 pp at the mean, and by 0.12 pp when attention is one standard deviation above. The channel operates in quiet periods and is muted in noisy ones.

Table 6: The framing effect at different levels of press attention

Term	$h = 6$	$h = 9$	$h = 12$	$h = 18$
Press framing, main effect	−0.189***	−0.345***	−0.400***	−0.385***
Framing $\times$ attention	+0.125	+0.198**	+0.281***	+0.288***
<i>Implied effect at $h = 12$:</i>				
Attention one s.d. below mean			−0.68	
Attention at mean			−0.40	
Attention one s.d. above mean			−0.12	

Note. Attention is the standardised Nexis count of inflation-related articles. n falls from 155 at $h = 6$ to 143 at $h = 18$. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Two readings fit the reversal. The first is saturation. When inflation is already

everywhere in the press, a month’s shift in framing is a small share of what any household encounters. Hence, the marginal article changes little, and the mechanism then operates on the margin of attention rather than on its level. The second is substitution. Months of heavy coverage are months of high inflation, when households observe prices in shops and have less need of a newspaper to tell them what is happening.

A persistent regressor entered as an interaction can produce a large coefficient by accident, so the result was checked by the randomisation procedure described in Section 5.3, with the placebo interacted with the actual attention series. The real interaction exceeds the placebo distribution with an empirical p -value of 0.006 at twelve months. Holding attention fixed makes this a harder test to pass, as the placebo inherits any spurious structure that the series carries.

The interaction implies that the channel should be visible in quiet periods and absent in loud ones, and the two inflation episodes in the sample illustrates this. Over the Global Financial Crisis window of 2007 to 2009, press framing moves expectations by -0.619 pp at nine months. The sign matches the full sample, and the magnitude is larger. The pattern from 2020 onwards is different. Press framing is followed by expectations 0.204 pp higher on impact, and 0.250 pp higher at three months, after which nothing is detectable at six, nine or twelve. A transmission channel produces a delayed response, as in Section 6.1. A response that appears immediately and disappears within a quarter is the signature of framing and expectations reacting to the same prices in the same month. The channel is not operating at all, and it is absent during the period when inflation coverage was heaviest.

6.5 Separating stance from information

Over the time window around each of 308 publication dates, a text surprise associated with an equity move of the opposite sign is a stance shock, while the one associated with a move of the same sign is an information shock. The unrotated surprise is null at every horizon, at $+0.083$ for RoBERTa tone and $+0.063$ for dictionary hawkishness at twelve months. Separating the two shocks changes this. The information shock is positive on both measures, reaching $+0.153$ for tone at twelve months and rising monotonically for the dictionary from $+0.154$ at ten months to $+0.311$ at seventeen. The stance shock is negative at most horizons and is not significant at any.

Table 7: Stance and information shocks, entered jointly

h	RoBERTa tone		Dictionary hawkishness	
	β^{MP}	β^{CBI}	β^{MP}	β^{CBI}
10	-0.071	+0.154**	-0.013	+0.154
12	-0.042	+0.153*	-0.030	+0.204*
15	-0.121	+0.110	-0.030	+0.262***
17	-0.082	+0.108	-0.016	+0.311***
<i>Unrotated surprise, same specification, $h = 12$:</i>				
	+0.083	($p = 0.176$)	+0.063	($p = 0.380$)
<i>Randomisation, empirical p at $h = 12$:</i>				
stance	0.759		0.939	
information	0.116		0.071	

Note. ε^{MP} is the stance shock and ε^{CBI} the central bank information shock, partitioned by the sign of the FTSE 100 response over the announcement window and entered jointly. Predicted signs $\beta^{MP} < 0$ and $\beta^{CBI} > 0$ were registered before estimation. $n = 229$ to 246. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

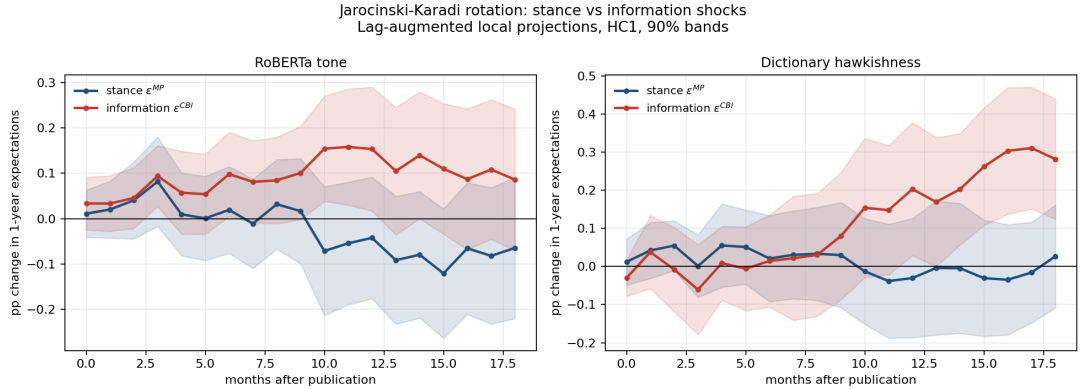


Figure 3: Stance and information shocks after the Jarociński-Karadi rotation. The unrotated surprise is null at every horizon, while the partition separates a positive information response from a weakly negative stance response. 90% bands.

$\beta^{CBI} > 0$ means that when communication reveals the economy is stronger than the public believed, household inflation expectations rise. This is the central bank information effect, and it is usually documented in markets and professional forecasters. The effect appears from ten months and grows afterwards, matching the delayed profile of Section 6.1 and consistent with diffusion but not an announcement effect. The randomisation procedure of Section 5.3 qualifies these estimates. Against 1,000 persistence-matched placebos at twelve months,

the stance shock returns empirical p -values of 0.759 for tone and 0.939 for the dictionary, which is close to noise. The information shock returns 0.116 and 0.071, which Section 5.3 treats as suggestive, and the information effect is reported on that footing.

6.6 The distribution of household beliefs

The monthly Citi/YouGov series used here reports only the the point estimate one-year-ahead household inflation expectation and provides no breakdown of responses, so it cannot show which parts of the distribution have shifted. The BoE/Ipsos Inflation Attitudes Survey instead publishes the share selecting each response band. These shares provide the binned distribution, while assigning midpoints recovers the whole distribution. The distribution clarifies the negative response reported in Section 6.1. Lower expectations follow household framing, and the distribution shows what that fall consists of.

Table 8: Press household framing and the Ipsos belief distribution

Dependent variable	$h = 4q$	$h = 6q$	$h = 8q$
Median expectation	-0.233	-0.307*	-0.348*
Share expecting 5% or more	-2.658	-3.517*	-4.614**
Share expecting below 2%	+2.894	+3.632	+4.725
Share answering ‘don’t know’	+0.478	+2.040***	+1.692***
Target credibility	+2.196*	+1.596	+2.646**
Disagreement (s.d. across bins)	-0.022	-0.038	-0.026
Citi/YouGov (benchmark)	-0.443*	-0.465*	-0.553***

Note. Quarterly. Shares are in percentage points of respondents; expectations in percentage points of inflation. $n = 47$. Lag-augmented local projections with HC1 standard errors. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Results show that the share of households with high inflation expectations falls. Eight quarters after a one-standard-deviation press framing surprise, the share of respondents expecting inflation of 5% or more falls by 4.6 percentage points, and by 3.5 points at six quarters. The share expecting less than 2% rises by 4.7 points over the same horizon, but that estimate is imprecise and significant at no horizon, so it carries no weight in either direction. What does rise precisely is the share answering that they do not know, by 2 points at six quarters and 1.7 at eight, and the share replying that the 2% target is about right, by 2.6 points at eight quarters. Disagreement, measured as the standard deviation across bins, does not move at any horizon.

A persuasion mechanism would imply a different distributional response. Persuading households that inflation will be low moves them out of a high band and into a low one, and if many are persuaded of the same thing, disagreement falls as beliefs converge. Neither occurs in the data. The estimated share in the low band does not increase significantly, and disagreement remains unchanged at every horizon. Instead, the data show a contraction in the alarmed tail, an increase in the share viewing the target as about right, and a statistically significant rise in ‘don’t know’ responses. Alarm falls and trust rises, while more households say they do not know what to expect, which is not what the transmission of a particular number would look like.

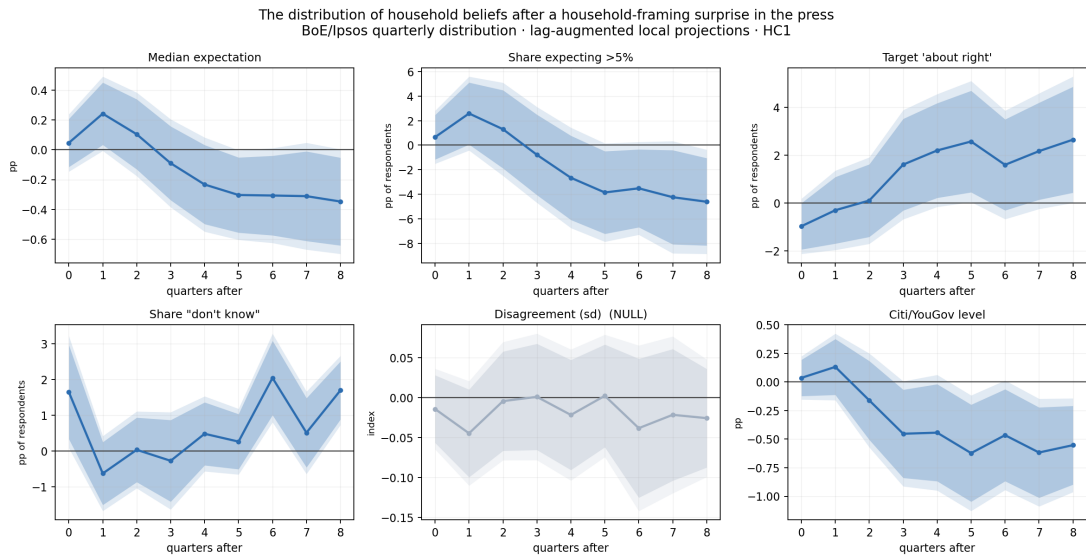


Figure 4: Household framing against each moment of the Ipsos quarterly belief distribution. Disagreement, the one null, is shown in grey.

The framing result already shows what the median leaves out. A median moves only when the household at the middle of the distribution changes bands, which makes it largely insensitive to shifts in the tails. At eight quarters, press framing moves the median expectation by 0.35 percentage points. Over the same horizon, 4.6% of respondents leave the band expecting inflation of 5% or more, and a further 1.7% report that they no longer know what to expect. Roughly one household in twenty leaves the band expecting 5% or more, while the median moves by a third of a percentage point.

Table 9: Identified shocks against the belief distribution

Dependent variable	Stance shock $h = 4q$	Information shock $h = 4q$	Policy shock $h = 4q$
Median expectation	−0.119**	+0.019	−0.118
Ipsos mean expectation	−0.098**	+0.009	−0.071
Share expecting 5% or more	−1.396**	+0.145	−1.652
Share expecting below 2%	+1.742**	−0.190	+0.798
Five-year share 5% or more	−1.139***	−0.116	−0.011
Disagreement	+0.012	−0.003	−0.039**
Citi/YouGov (benchmark)	−0.075	+0.072	+0.058
n	76	76	28

Note. Quarterly, at a four-quarter horizon. Shares are in percentage points of respondents. The stance and information shocks are those of Section 6.5; the policy shock is the residual of equation (5), standardised. Disagreement responses to the policy shock are -0.027^* at two quarters and -0.086^{**} at eight. The final row is the mean-based benchmark: the object a first-moment design observes. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The stance shock is the clearer case. Four quarters after a one-standard-deviation stance shock, the share expecting 5% or more falls by 1.4 points and the share expecting less than 2% rises by 1.7, both significant at 5%. The Ipsos mean over the same horizon moves by 0.098 pp and the median by 0.119. On the Citi series, the shock is close to zero at any horizon. The small mean is driven by the two-tail movements, which run in opposite directions and largely cancel. The stance shock also moves the five-year share expected to be 5% or more by 1.1 points at four quarters, where household framing does nothing at any horizon. The two channels influence different parts of the term structure of beliefs. Stance reaches long-run expectations, and framing does not.

Following Romer and Romer (2004), a monetary policy shock is identified as the residual from regressing the change in Bank Rate on the Committee’s own published projections, their revisions, and the pre-meeting rate level from 2013 to 2026. The projections explain about 90% of quarterly rate changes, leaving a residual standard deviation of 0.106 pp. The shock leaves the Citi series unmoved at every horizon. It compresses disagreement (a dispersion measure of the survey) by 0.039 at four quarters and 0.086 at eight, and it is the only shock tested that does so. A rate surprise resolves uncertainty about the path of policy and beliefs converge. Three shocks that are negligible in a mean-based series influence the distributional series: framing shrinks the alarmed tail and raises the share

declining to answer, stance redistributes between both tails and reaches long-run beliefs, and policy compresses disagreement.

7 Discussion

7.1 The results in light of Carroll (2003)

What the Bank says about interest rates may not move household expectations, but what it says about households appears to do so. Coverage framed around household budgets is followed by lower inflation expectations. Three features of that finding require explanation: its sign, its timing, and the conditions under which it disappears. This section sets them against the model that motivated the design.

Carroll (2003) proposes that households do not observe the professional forecast directly. They learn them from news, and in any period each individual faces a probability λ of absorbing the content of an article on inflation. Those who do not encounter one continue to hold the last forecast they read. Carroll’s framework is similar to Calvo (1983) sticky prices, with λ playing the role of the price-resetting probability. The aggregate household expectation is

$$M_t = \lambda S_t + (1 - \lambda)M_{t-1}, \quad (6)$$

where M is the household (Michigan Survey) expectation and S the professional (SPF) forecast. Carroll estimates $\lambda = 0.273$ per quarter, meaning roughly a quarter of households absorb the current forecast in a given quarter.

Coverage matters in both analyses. Carroll regresses the squared distance between household and professional forecasts on an index of the intensity of inflation coverage, finding a coefficient of -1.72 over 1981-2000, so household expectations are closer to professional ones when coverage is heavy. Splitting the sample on coverage, the absorption rate rises from $\lambda = 0.210$ in periods of light coverage to $\lambda = 0.699$ in periods of heavy coverage. Section 6.4 finds the opposite: the framing effect falls from 0.68 pp when press attention is a standard deviation below its mean to 0.12 pp a standard deviation above.

Two findings are consistent with Carroll’s results. The first is that Carroll asks whether the level of the professional forecast reaches households faster under heavy coverage, whereas the interaction here asks whether a marginal shift in framing carries more weight. The second is that attention in this sample is concentrated in 2022 and 2023, so the interaction may be capturing a regime change. Coibion and Gorodnichenko (2026) offer a third interpretation. They describe a cycle of selective inattention, in which households attend to inflation only when it is high and monetary policy appears to be failing, so that communication is most powerful when inflation is quiet and least effective when it is not. Section 6.4

explores this pattern.

Households rely on forecasts received at different dates, so disagreement should persist, while a signal reaching many households at once should reduce it, which is the third implication of Carroll. Testing this requires the distribution, which the BoE/Ipsos bands provide. Disagreement, measured as the standard deviation across bins, does not move for press framing at any horizon, nor for Bank framing, the stance shock or the information shock. Text appears to move where the mass of the distribution sits without changing how far apart households are.

7.2 Relation to Tong and Cherry (2026)

Tong and Cherry (2026) is the closest existing study to the results in Section 6.5, with three points of similarity. The shocks are constructed from central bank text and from newspaper narratives, comparing each central bank narrative against the pre-announcement press coverage to obtain a surprise. They use co-movement with daily equity prices to separate those narrative surprises into stance and information components. They then estimate the response of one-year-ahead household inflation expectations to each component separately. Built on a different corpus, a different window, and an independently constructed measure, the results here are comparable in sign and ordering. Tightening stance communication lowers expectations, expansionary information communication raises them, and the unrotated surprise recovers neither. The newspapers there supply the benchmark against which the Bank’s language is measured, whereas the press measure in Section 4.3 is a channel in its own right.

Their study argues that conventional shocks identified from high-frequency asset-price movements do not produce these responses. Section 6.6 reaches the same conclusion by a different route, since the Romer and Romer policy shock identified from the Committee’s own published projections leaves the mean household expectation unmoved at every horizon.

The two analyses suggest different roles for attention. Tong and Cherry find the response larger and earlier when attention is high, across the Bank of Canada, the Bank of England and the Federal Reserve, whereas Section 6.4 finds the framing effect falling from 0.68 pp at low attention to 0.12 pp at high. Tong and Cherry note that the theoretical effect of attention is ambiguous, as greater attention may make households more likely to notice central bank messages, or it may rise precisely when inflation is high and policy appears to be failing, weakening the credibility of what policymakers say.

7.3 Relation to other work

Three other studies share elements with this work’s design. Coibion, Gorodnichenko and Weber (2022) examine how household inflation expectations respond to exposure to central bank messages. This dissertation instead shows that households encounter the Bank’s language as the press reports it, and Section 6.1 finds the press measure acting later and persisting longer than the Bank measure. Lamla and Maag (2012) study the role of media for inflation forecast disagreement between households and forecasters, which is the closest existing work to the third implication tested in Section 7.1. The evidence is that text does not move dispersion at all: neither framing measure, nor the stance shock, nor the information shock shifts the standard deviation across bands at any horizon.

Adrian, Boyarchenko and Giannone (2019) establish the approach for output growth, finding the lower quantiles move with financial conditions while the upper quantiles stay flat, and López-Salido and Loria (2024) apply quantile regression to inflation across a panel of OECD countries, finding that the conditional mean was recently low and stable while the tails varied considerably. Section 6.6 asks that question of household beliefs, and finds that a stance shock moving the Ipsos mean by a tenth of a percentage point simultaneously shifts 1.4 points out of the band expecting 5% or more and 1.7 points into the band expecting below 2%.

8 Conclusion

This dissertation asked whom central bank communication is addressed instead of what direction it points in. Results show that household framing moves expectations. Household-framed press coverage is followed by inflation expectations lower by 0.35 percentage points at twelve months, and Bank text gives -0.26 , on a corpus whose framing measure shows no correlation with the first. No detectable response appears before the communication, and the effect begins to build only from the sixth month. This delay is consistent with Carroll (2003), while the absence of earlier movement argues against communication and expectations simply moving together. Rebuilding the press measure with a classifier (Central-BankRoBERTa) that uses no word list returns -0.347 against -0.346 at the same horizon.

The direction measures behave differently. Pooled, they leave household expectations unmoved, and the rotation shows why: the surprise mixes a stance shock and an information shock of opposite sign, and separating them recovers an information effect of 0.15 to 0.31 percentage points. Direction reaches policy and markets instead of households, with Bank tone forecasting the policy rate to eighteen months, and the press carries little of it.

The distribution reveals what the mean does not. A stance shock moves 1.4 points out of the band expecting 5% or more and 1.7 points into the band expecting below 2% four quarters out. Household framing shrinks the alarmed tail by 4.6 points at eight quarters and raises the share replying that they do not know, while disagreement does not move at any horizon or under any shock. The distribution shifts without spreading.

The channel is state-dependent. Framing moves expectations by 0.68 percentage points when press attention is a standard deviation below its mean and by 0.12 when it is a standard deviation above, where the effect is not statistically significant. Communication works when coverage is ordinary and stops working when coverage is heavy. The window in which the Bank can shape household expectations is therefore widest when inflation is not the story, and closes as it becomes one.

Three limitations arise from how the communication measures are constructed and identified. First, the classifier is fine-tuned on sentences hand-labelled from Federal Reserve speeches, using positive and negative, not hawkish and dovish labels. It therefore captures how the Bank sounds, not what the Committee intends, while applying it to Bank of England documents and newspaper prose extends it beyond

its training corpus. Second, the rotation classifies announcements through the co-movement of text and equity prices. This separates a stance shock from an information shock only if these are the only two shocks moving both series within the window. Any other effect on text and equities over the same days is absorbed into one of the two, potentially distorting the identified shocks. Third, the press measure captures how the sampled articles frame inflation, not what households read. Nexis limits how much material can be collected, so the corpus uses a sample drawn from five major newspapers. Without this constraint, the measure could score every article published by those newspapers during the period, providing a more complete record of press framing.

The findings of this study open several directions for future research. A classifier fine-tuned on hand-labelled Bank of England sentences would remove the extrapolation, following Nițoi, Pochea and Radu (2023), who hand-annotate sentences from central bank communications, fine-tune and validate against an external benchmark. A third high-frequency series, sterling or the inflation swap curve, would overidentify the rotation and make the two-shock restriction testable. The largest gain would come from the respondent-level microdata behind the Bank’s own survey. Estimating quantiles from the answers themselves would replace tails inferred from bands with tails measured directly. The demographic information collected by Ipsos would also extend the analysis from comparisons across quarters to comparisons among respondents within the same quarter.

Although this study includes episode analysis, its conclusions rest on windows too short to carry much weight: the financial crisis window leaves 24 observations at nine months and 21 at twelve, and the response is not significant in most of the regressions. Two hand-picked episodes are also two draws, chosen because they are known to be episodes rather than by a rule applied to the data. A state-dependent local projection with a smooth transition between regimes would use the whole sample instead, estimating how the response varies with inflation or with attention rather than imposing a break at dates fixed in advance.

Bibliography

- Adrian, T., Boyarchenko, N. and Giannone, D. (2019) ‘Vulnerable growth’, *American Economic Review*, 109(4), pp. 1263–1289. doi: 10.1257/aer.20161923.
- Apel, M., Blix Grimaldi, M. and Hull, I. (2022) ‘How much information do monetary policy committees disclose? Evidence from the FOMC’s minutes and transcripts’, *Journal of Money, Credit and Banking*, 54(5), pp. 1459–1490. doi: 10.1111/jmcb.12885.
- Apel, M. and Blix Grimaldi, M. (2012) ‘The information content of central bank minutes’, *Sveriges Riksbank Working Paper Series* No. 261.
- Ash, E. and Hansen, S. (2023) ‘Text algorithms in economics’, *Annual Review of Economics*, 15, pp. 659–688. doi: 10.1146/annurev-economics-082222-074352.
- Baker, S. R., Bloom, N. and Davis, S. J. (2016) ‘Measuring economic policy uncertainty’, *Quarterly Journal of Economics*, 131(4), pp. 1593–1636. doi: 10.1093/qje/qjw024.
- Bertrand, M., Duflo, E. and Mullainathan, S. (2004) ‘How much should we trust differences-in-differences estimates?’, *Quarterly Journal of Economics*, 119(1), pp. 249–275. doi: 10.1162/003355304772839588.
- Blinder, A. S., Ehrmann, M., Fratzscher, M., de Haan, J. and Jansen, D.-J. (2008) ‘Central bank communication and monetary policy: a survey of theory and evidence’, *Journal of Economic Literature*, 46(4), pp. 910–945. doi: 10.1257/jel.46.4.910.
- Blinder, A. S., Ehrmann, M., de Haan, J. and Jansen, D.-J. (2024) ‘Central bank communication with the general public: promise or false hope?’, *Journal of Economic Literature*, 62(2), pp. 425–457. doi: 10.1257/jel.20231683.
- Calvo, G. A. (1983) ‘Staggered contracts in a utility-maximizing framework’, *Journal of Monetary Economics*, 12(3), pp. 383–398. doi: 10.1016/0304-3932(83)90060-0.
- Carroll, C. D. (2003) ‘Macroeconomic expectations of households and professional forecasters’, *Quarterly Journal of Economics*, 118(1), pp. 269–298. doi: 10.1162/00335530360535207.
- Coibion, O. and Gorodnichenko, Y. (2026) ‘Inflation expectations and monetary

- policy: what have we learned and to what end?', *The Econometrics Journal*, utag004. doi: 10.1093/ectj/utag004.
- Coibion, O., Gorodnichenko, Y. and Kamdar, R. (2018) 'The formation of expectations, inflation, and the Phillips curve', *Journal of Economic Literature*, 56(4), pp. 1447–1491. doi: 10.1257/jel.20171300.
- Coibion, O., Gorodnichenko, Y. and Weber, M. (2022) 'Monetary policy communications and their effects on household inflation expectations', *Journal of Political Economy*, 130(6), pp. 1537–1584. doi: 10.1086/718982.
- Gürkaynak, R. S., Sack, B. and Swanson, E. T. (2005) 'Do actions speak louder than words? The response of asset prices to monetary policy actions and statements', *International Journal of Central Banking*, 1(1), pp. 55–93.
- Hansen, S., McMahon, M. and Tong, M. (2019) 'The long-run information effect of central bank communication', *Journal of Monetary Economics*, 108, pp. 185–202. doi: 10.1016/j.jmoneco.2019.09.002.
- Jarociński, M. and Karadi, P. (2020) 'Deconstructing monetary policy surprises: the role of information shocks', *American Economic Journal: Macroeconomics*, 12(2), pp. 1–43. doi: 10.1257/mac.20180090.
- Jordà, Ò. (2005) 'Estimation and inference of impulse responses by local projections', *American Economic Review*, 95(1), pp. 161–182. doi: 10.1257/0002828053828518.
- Kalamara, E., Turrell, A., Redl, C., Kapetanios, G. and Kapadia, S. (2022) 'Making text count: economic forecasting using newspaper text', *Journal of Applied Econometrics*, 37(5), pp. 896–919. doi: 10.1002/jae.2907.
- Kuttner, K. N. (2001) 'Monetary policy surprises and interest rates: evidence from the Fed funds futures market', *Journal of Monetary Economics*, 47(3), pp. 523–544. doi: 10.1016/S0304-3932(01)00055-1.
- Lamla, M. J. and Maag, T. (2012) 'The role of media for inflation forecast disagreement of households and professional forecasters', *Journal of Money, Credit and Banking*, 44(7), pp. 1325–1350. doi: 10.1111/j.1538-4616.2012.00534.x.
- Larsen, V. H., Thorsrud, L. A. and Zhulanova, J. (2021) 'News-driven inflation expectations and information rigidities', *Journal of Monetary Economics*, 117,

- pp. 507–520. doi: 10.1016/j.jmoneco.2020.03.004.
- López-Salido, D. and Loria, F. (2024) ‘Inflation at risk’, *Journal of Monetary Economics*, 145. doi: 10.1016/j.jmoneco.2024.103570.
- Loughran, T. and McDonald, B. (2011) ‘When is a liability not a liability? Textual analysis, dictionaries, and 10-Ks’, *Journal of Finance*, 66(1), pp. 35–65. doi: 10.1111/j.1540-6261.2010.01625.x.
- Montiel Olea, J. L. and Plagborg-Møller, M. (2021) ‘Local projection inference is simpler and more robust than you think’, *Econometrica*, 89(4), pp. 1789–1823. doi: 10.3982/ECTA18756.
- Nițoi, M., Pochea, M.-M. and Radu, Ș.-C. (2023) ‘Unveiling the sentiment behind central bank narratives: a novel deep learning index’, *Journal of Behavioral and Experimental Finance*, 38, 100809. doi: 10.1016/j.jbef.2023.100809.
- Pfeifer, M. and Marohl, V. P. (2023) ‘CentralBankRoBERTa: a fine-tuned large language model for central bank communications’, *Journal of Finance and Data Science*, 9. doi: 10.1016/j.jfds.2023.100114.
- Romer, C. D. and Romer, D. H. (2004) ‘A new measure of monetary shocks: derivation and implications’, *American Economic Review*, 94(4), pp. 1055–1084. doi: 10.1257/0002828042002651.
- Tong, E. and Cherry, R. (2026) ‘Central bank communications that reach the public’, *Bank of England Staff Working Paper* No. 1,194.
- Woodford, M. (2005) ‘Central bank communication and policy effectiveness’, in *The Greenspan Era: Lessons for the Future*. Proceedings of the Economic Policy Symposium, Jackson Hole. Federal Reserve Bank of Kansas City, pp. 399–474.

Appendix 1: Dictionaries and word lists

A1.1 Household and market frame vocabularies

Household frame (54 terms). household, households, family, families, shopper, shoppers, cost of living, living costs, weekly shop, grocery, groceries, supermarket, wage, wages, pay packet, earnings, salaries, mortgage, mortgages, rent, rents, tenant, tenants, energy bill, energy bills, gas bill, fuel bill, utility bill, bills, food price, food prices, petrol, budget squeeze, squeezed, hard-pressed, breadline, poverty, pensioner, pensioners, saver, savers, borrower, borrowers, benefit, benefits, universal credit, afford, affordability, unaffordable, basket, weekly budget, home owner, homeowners, first-time buyer.

Market frame (43 terms). investor, investors, trader, traders, market, markets, gilt, gilts, bond, bonds, yield, yields, equity, equities, share price, shares, stock market, ftse, hedge fund, asset manager, fund manager, portfolio, bank profit, profits, margin, margins, earnings season, corporate, firms, businesses, industry, manufacturer, manufacturers, balance sheet, liquidity, leverage, swap, swaps, derivative, derivatives, city of london, analysts, economists at.

A1.2 The hawkishness dictionary

Persistence (18). persistent, persistence, persisting, persistently, stubborn, stubbornly, sticky, stickier, stickiness, entrenched, embedded, enduring, lasting, long-lasting, prolonged, protracted, lingering, ingrained.

Level (14). elevated, above-target, rampant, intense, acute, severe, extreme, pronounced, heightened, unsustainable, unsustainably, overshoot, overshooting, resurgence.

Dynamics (11). second-round, wage-price, wage-setting, price-setting, self-fulfilling, spiral, spirals, spiralling, stagflation, stagflationary, stoking.

Policy response (16). tightening, tighten, tighter, restrictive, restraining, contractionary, aggressive, aggressively, forceful, forcefully, vigilant, wary, attentive, hawkish, hiking, taming.

Cost seriousness (9). painful, damaging, costly, harsh, serious, threat, uncomfortable, worsening, abrupt.

Terms of the opposite sign were kept out and terms with no clear directional reading were dropped (27). And five terms survived to a late stage before being removed.

Table 10: Candidates removed from the hawkishness dictionary

Term	BoE frequency	Press frequency
catch-up	26	34
feeding	109	137
scarring	26	37
high	2,208	7,802
higher	8,722	10,373

Appendix 2: Variables and data construction

A2.1 Variables, definitions and sources

Table 11: Household inflation expectations

Variable	Definition	Frequency	Source
citi_yougov_1y	Median expectation of inflation over the next 12 months	Monthly	Citi/YouGov
citi_yougov_5y10y	Expectation of inflation 5 to 10 years ahead	Monthly	Citi/YouGov
next12m_median	Median of the binned 12-month expectation	Quarterly	BoE/Ipsos
next12m_mean	Mean of the binned 12-month expectation, bin midpoints	Quarterly	BoE/Ipsos
next12m_disagreement	Standard deviation across bins	Quarterly	BoE/Ipsos
next12m_share_5plus	Share of respondents expecting 5% or more	Quarterly	BoE/Ipsos
next12m_share_below2	Share expecting less than 2%	Quarterly	BoE/Ipsos
next12m_share_dontknow	Share answering ‘don’t know’	Quarterly	BoE/Ipsos
next5y_share_5plus	Share expecting 5% or more over five years	Quarterly	BoE/Ipsos
target_aboutright	Share replying that the 2% target is about right	Quarterly	BoE/Ipsos

Table 12: Macroeconomic series

Variable	Definition	Frequency	Source
cpi	CPI inflation, 12-month rate, not an index	Monthly	ONS
cpi_food_yoy	Food and non-alcoholic beverages, 12-month rate	Monthly	ONS
cpi_energy_yoy	Energy component, 12-month rate	Monthly	ONS
cpi_core_yoy	Core CPI, 12-month rate	Monthly	ONS
unemployment_rate	LFS unemployment rate, 16+	Monthly	ONS
gdp_monthly_index	Monthly GDP index	Monthly	ONS
bank_rate	Bank Rate, end of month	Monthly	Bank of England
brent_usd_bbl	Brent crude, USD per barrel	Monthly	LSEG
nbp_gas_gbp_therm	NBP natural gas, pence per therm	Monthly	LSEG
sterling_eri	Sterling effective exchange rate index	Monthly	Bank of England

Table 13: Market series

Variable	Definition	Frequency	Source
infl_spot_3y ... infl_spot_20y infl_5y5y_fwd	BoE implied inflation curve at 3, 5, 10, 15 and 20 years Five-year five-year forward inflation	Monthly Monthly	Bank of England Bank of England
breakeven_5y, breakeven_10y	Index-linked gilt breakevens	Monthly	LSEG
swap_1y ... swap_ 10y	Inflation swap rates	Monthly	LSEG
ftse100	FTSE 100 close, 5,463 trading days	Daily	Investing.com

Table 14: Text and attention measures

Variable	Definition	Frequency	Source
hawk_boe	Dictionary hawkishness, share of stems in BoE documents	Monthly	Own construction
cb_positive	Share of BoE sentences classified positive	Monthly	CentralBank- RoBERTa
cb_households	Share of BoE sentences classified as concerning households	Monthly	CentralBank- RoBERTa
news_hh_focus	Household-frame terms as a share of household plus market terms	Monthly	Own construction, Nexis
attention_nexis	Count of inflation-related articles	Monthly	Nexis
gt_inflation	Google Trends index, UK, ‘inflation’	Monthly	Google Trends
minutes_fog, minutes_fk	Gunning fog and Flesch–Kincaid readability	Monthly	Own construction

Table 15: Policy and forecast series

Variable	Definition	Frequency	Source
dissent_share	Share of MPC members voting against the majority	Per meeting	Bank of England
cpi_h0, cpi_h1, cpi_h2	MPC modal CPI projection at the current, one- and two-year horizons	Per Report	Monetary Policy Report
gdp_h0, gdp_h1, gdp_h2	MPC modal GDP growth projection	Per Report	Monetary Policy Report
cpi_yy_poll_ median	Median forecaster expectation of CPI	Monthly	Reuters poll
gfk_consumer_ confidence	GfK consumer confidence barometer	Monthly	GfK

A2.2 Construction of the newspaper corpus

Table 16: Newspaper corpus construction, 172 months

	Articles
Distinct articles in the population	130,802
Retrieved and parsed	11,192
Removed as near-duplicates	2,489
Removed by the uniform sampling rule	2,077
Retained in the analysis corpus	6,625
	Sampling rate
Mean before the rule	7.2%
Standard deviation before the rule	0.021
Mean after the rule	5.3%
Standard deviation after the rule	0.010

Note. Imposing the rule halves the cross-month dispersion of the sampling rate. Twenty-four months fall short of their target, twenty because the article volume exceeded what could be retrieved and four because fewer than twenty-five distinct articles were available. The month-level audit is in the replication materials.

A2.3 Text preparation

Bank of England documents were downloaded as PDFs, converted to text, and stripped of headers, footers, standard annexes and bare URLs by a fixed list of patterns, with whitespace tidied and paragraph structure preserved. Newspaper articles were parsed from Nexis exports into title, date, source and body.

Both corpora then pass through the same tokeniser. Text is lower-cased, decimal points are protected so that figures are not split at the point, the text is divided into sentences at full stops, question marks and exclamation marks, and each sentence is reduced to its alphabetic tokens, keeping hyphenated forms and discarding numbers, punctuation and single characters. Sentences of fewer than four tokens are dropped. No stop-word list is applied, since every measure is a share computed within sentences already restricted by topic.

The word2vec model of Section 4.1 is trained on the combined corpora at 100 dimensions, with a window of five words, a minimum frequency of twenty, ten epochs and the random seed fixed at 42. Twenty-five words naming the outcome or the instrument, among them inflation, CPI, RPI, price, rate, Bank, Committee, MPC, policy and monetary, are blocked from the candidate list so that the dictionary cannot select the quantity it is meant to explain. Porter stemming is applied when documents are scored and not before training, so that the embedding learns from inflected text while a document is matched on inflected forms as well as on the exact dictionary word.

A2.4 How the data were assembled

The project runs as an ordered pipeline of Python scripts, one folder per stage. `01_download` retrieves the Monetary Policy Committee minutes and the Monetary Policy Reports as PDFs from the Bank’s website, and the macroeconomic series from the Office for National Statistics and the Bank’s statistical database. `02_clean` converts each source to a single file at monthly frequency. `03_text_processing` extracts and cleans the text of both corpora, trains the word2vec model, builds the stance dictionary and scores every document. `04_news` parses the Nexis exports, applies the sampling rule of Section 3.3, and builds the attention and duplication series. `04_merge` joins every cleaned file onto one monthly grid and writes the master panel. `05_analysis` estimates every specification reported in Sections 6 and 7 and writes the tables of Appendix 3. `06_figures_tables` regenerates Appendix 1 from the word lists themselves, so that the printed dictionary cannot drift from the one used in estimation.

Two steps sit outside that sequence. The CentralBankRoBERTa classifiers of Section 4.2 require a GPU and were run on Google Colab, with the scored sentences returned to the pipeline and merged. The newspaper material was collected through the Nexis web interface, since no programmatic access was available, which is what limits the size of the corpus.

Seeds are fixed wherever a result depends on one: 42 for the word2vec model of Section 4.1, 20260822 for the randomisation test of Section 5.3, and 0 for the cross-validation of Section 6.2.

The replication materials contain the code, the cleaned monthly series, the master panel and the month-level sampling audit. They do not contain the newspaper article texts or the LSEG series, which are licensed and cannot be redistributed. The Bank of England documents are public, and the download script rebuilds that corpus from source.

Appendix 3: Additional results

This appendix reports the estimates behind the figures and the summary tables in the text: every stage of the estimation, the local projections at all horizons, the lead placebos, the two constructions of the press measure, the episode splits, the distributional results, the decomposition of press framing into its transmitted and autonomous parts, the high-frequency event study, the reported nulls, and the diagnostics. Variable names are given as they appear in the estimation code, and throughout this appendix *** denotes $p < 0.01$, ** denotes $p < 0.05$ and * denotes $p < 0.10$.

A3.1 Full results, stage by stage

Table 17: Stage 1 — first-stage fit and the share of tone that is a surprise

tone measure	R ²	surprise share	n
hawk_boe	0.224	77.6%	215
cb_tone	0.166	83.4%	215

Table 18: Stage 1 — first-stage coefficients, **cb_tone**

regressor	coefficient	t
Intercept	−0.5496	−6.84
cpi	−0.0653	−1.57
bank_rate	−0.0139	−3.67
infl_5y5y_fwd	+0.0090	+0.36
nbp_gas_gbp_therm	−0.0002	−1.23
cpi_yy_poll_median	+0.0540	+1.28

Table 19: Stage 2 — validation: does the measure track what the MPC did? This is the source of the third validation test in Section 4.4: the dictionary’s +0.027 correlation with the next rate change and its 5.6× drift, against the transformer’s 1.26×.

measure	corr. next rate change	corr. CPI	2005–07 mean	2024–26 mean	drift
hawk_boe	+0.027	+0.381	+0.241	+1.352	5.61×
cb_tone	+0.158	−0.349	−0.560	−0.703	1.26×

Table 20: Stage 3 — episode comparison: hawk_boe.

episode	β	se	t	p
Full sample	-0.0683	0.0709	-0.96	0.336
GFC	—	—	—	—
ZLB, CPI 5%+	-0.2611	0.1499	-1.74	0.093 *
Taper/quiet	+0.1821	0.1543	+1.18	0.244
COVID surge	+0.2713	0.2405	+1.13	0.273
Post-peak	-0.3272	0.1624	-2.01	0.056 *

Table 21: Stage 3 — episode comparison: cb_tone

episode	β	se	t	p
Full sample	+0.3351	0.2442	+1.37	0.172
GFC	—	—	—	—
ZLB, CPI 5%+	+0.0897	0.3542	+0.25	0.802
Taper/quiet	+0.0074	0.1669	+0.04	0.965
COVID surge	-0.7649	0.3810	-2.01	0.059 *
Post-peak	+0.1141	0.5957	+0.19	0.850

Table 22: Stage 4 — local projections (Jordà 2005): hawk_boe, full sample

h	β	se	t	p	n
0	-0.0683	0.1024	-0.67	0.505	206
1	-0.0239	0.1238	-0.19	0.847	206
2	-0.0263	0.1098	-0.24	0.811	205
3	-0.0537	0.1512	-0.36	0.723	205
4	+0.0442	0.1685	+0.26	0.794	204
5	-0.0136	0.1801	-0.08	0.940	203
6	-0.0144	0.1893	-0.08	0.939	202
7	+0.0186	0.2021	+0.09	0.927	202
8	+0.0007	0.2448	+0.00	0.998	201
9	+0.0800	0.2705	+0.30	0.768	200
10	+0.0611	0.3060	+0.20	0.842	200
11	-0.0256	0.3503	-0.07	0.942	199
12	+0.0710	0.3929	+0.18	0.857	198

Table 23: Stage 4 — local projections (Jordà 2005): **hawk_boe**, moving regime

h	β	se	t	p	n
0	+0.0741	0.1661	+0.45	0.657	94
1	+0.1259	0.2234	+0.56	0.574	94
2	+0.1941	0.2148	+0.90	0.368	93
3	+0.0587	0.2887	+0.20	0.839	93
4	+0.1609	0.2741	+0.59	0.559	92
5	+0.3393	0.3315	+1.02	0.309	91
6	+0.4485	0.3926	+1.14	0.257	90
7	+0.5031	0.3794	+1.33	0.188	90
8	+0.6186	0.4663	+1.33	0.188	89
9	+0.6023	0.4672	+1.29	0.201	89
10	+0.7996	0.5210	+1.53	0.129	88
11	+0.7591	0.5388	+1.41	0.163	87
12	+0.5684	0.4855	+1.17	0.245	87

Table 24: Stage 4 — local projections (Jordà 2005): **cb_tone**, full sample

h	β	se	t	p	n
0	+0.3351	0.2990	+1.12	0.264	206
1	+0.8533	0.4413	+1.93	0.055 *	206
2	+1.0952	0.6795	+1.61	0.109	205
3	+1.3658	0.8753	+1.56	0.120	205
4	+1.1667	1.0361	+1.13	0.261	204
5	+1.2342	1.0722	+1.15	0.251	203
6	+1.5650	1.0987	+1.42	0.156	202
7	+1.7446	1.0984	+1.59	0.114	202
8	+1.9255	1.1415	+1.69	0.093 *	201
9	+1.9824	1.2532	+1.58	0.115	200
10	+2.3089	1.4402	+1.60	0.111	200
11	+2.3706	1.5761	+1.50	0.134	199
12	+2.1873	1.6003	+1.37	0.173	198

Table 25: Stage 4 — local projections (Jordà 2005): `cb_tone`, moving regime

h	β	se	t	p	n
0	+0.5707	0.4764	+1.20	0.234	94
1	+1.5661	0.8482	+1.85	0.068 *	94
2	+2.6127	1.3364	+1.96	0.054 *	93
3	+3.7245	1.6704	+2.23	0.028 **	93
4	+4.1713	1.8688	+2.23	0.028 **	92
5	+4.0279	1.9419	+2.07	0.041 **	91
6	+4.4927	2.0170	+2.23	0.029 **	90
7	+4.7809	2.0397	+2.34	0.021 **	90
8	+4.6679	2.1533	+2.17	0.033 **	89
9	+4.7760	2.2533	+2.12	0.037 **	89
10	+5.1253	2.1337	+2.40	0.019 **	88
11	+5.0049	2.1729	+2.30	0.024 **	87
12	+4.7877	2.0883	+2.29	0.024 **	87

Table 26: Stage 5 — the placebo fails: the surprise ‘predicts’ expectations before it occurred

specification	horizon	β	se	t	p
baseline	3 months before	+1.0149	0.5708	+1.78	0.079 *
baseline	6 months before	+2.7094	1.0472	+2.59	0.011 **

Table 27: Stage 5 — fix (a): controlling for the six-month pre-trend. The effect survives

h	β	se	t	p	n
3	+3.7575	1.5531	+2.42	0.018 **	93
6	+4.6351	1.9109	+2.43	0.017 **	90
10	+4.6486	1.8603	+2.50	0.014 **	88
12	+4.2464	1.8186	+2.33	0.022 **	87

Table 28: Stage 5 — fix (b): first-differencing the surprise. The effect vanishes

h	β	se	t	p	n
3	+0.6631	0.7721	+0.86	0.394	72
6	+0.2582	0.8985	+0.29	0.775	70
10	+0.4226	0.8126	+0.52	0.605	69
12	+0.3139	0.8069	+0.39	0.699	69

Table 29: Stage 5 — quiet-regime placebo, where expectations are flat by construction

h	β	t	p	n
3	+0.0831	+0.11	0.912	99
6	−0.7181	−0.75	0.456	97
10	−1.9990	−1.32	0.191	94
12	−2.3133	−1.84	0.069 *	93

Table 30: Stage 6 — robustness of the headline specification (h = 10, moving regime)

variation	β	se	t	p	n
baseline	+5.1253	2.1337	+2.40	0.019 **	88
no controls	+4.0344	1.9689	+2.05	0.044 **	88
+ lagged level of DV	+3.9779	2.1821	+1.82	0.072 *	88
+ 6-month pre-trend	+4.6486	1.8603	+2.50	0.014 **	88
drop COVID 2020–21	+4.0071	1.9787	+2.03	0.047 **	72
drop hiking 2022–23	+5.3673	2.4381	+2.20	0.031 **	76
pre–2020 only	+3.6009	2.2010	+1.64	0.108	53
first-differenced surprise	+0.4226	0.8126	+0.52	0.605	69

Table 31: Stage 7 — mechanism (Carroll’s second implication)

sample	tone	interaction β	t	p	n
full	hawk_boe	+0.0249	+0.18	0.859	206
moving	hawk_boe	+0.0272	+0.13	0.896	95
full	cb_tone	−0.3162	−1.00	0.321	206
moving	cb_tone	−0.9357	−1.38	0.170	95

Table 32: Stage 9 — market validation. A measure of central bank communication should move the prices of instruments whose value depends on future inflation. Each row regresses a monthly change in a market-implied inflation series (one- and five-year swap rates, the five-year breakeven, and the five-year five-year forward) on a tone surprise. Nothing moves decisively. The test is run at monthly frequency, however, and markets reprice within minutes of an announcement, so a month-long window buries the response under twenty days of unrelated news. The daily-frequency version is in A3.8, and it does find an effect.

dependent variable	tone	β	t	p	n
swap_1y	hawk_boe	-0.0562	-0.60	0.549	189
swap_1y	cb_tone	+0.3375	+1.18	0.239	189
swap_5y	hawk_boe	-0.0470	-0.93	0.355	189
swap_5y	cb_tone	+0.3853	+1.84	0.067 *	189
breakeven_5y	hawk_boe	-0.0990	-1.25	0.212	109
breakeven_5y	cb_tone	+0.1946	+0.83	0.407	109
infl_5y5y_fwd	hawk_boe	-0.0330	-1.63	0.105	215
infl_5y5y_fwd	cb_tone	+0.1856	+2.16	0.032 **	215

A3.2 Lead placebos

Table 33: Part A — lead placebos on the headline specification: Press household framing

horizon	beta	s.e.	p	n
h = -6 (lead)	+0.019	0.051	0.712	156
h = -3 (lead)	+0.035	0.030	0.235	159
h = 12 (headline)	-0.346***	0.083	0.000	149

Table 34: Part A — lead placebos on the headline specification: BoE household framing

horizon	beta	s.e.	p	n
h = -6 (lead)	-0.019	0.054	0.722	158
h = -3 (lead)	+0.001	0.029	0.985	161
h = 12 (headline)	-0.255***	0.084	0.002	158

A3.3 The press measure built two ways

Table 35: Agreement between the two series

comparison	correlation	n
raw series, levels	0.794	172
raw series, first differences	0.724	171
orthogonalised surprises	0.793	168

Table 36: The same impulse response, both measures

h	keyword ratio	transformer classifier	n
0	+0.063**	−0.011	160
3	+0.023	−0.015	158
6	−0.128*	−0.165**	155
9	−0.284***	−0.296***	152
12	−0.346***	−0.347***	149
18	−0.378***	−0.350***	143

A3.4 Episode estimates

Table 37: GFC (2007–01 to 2009–12)

regressor	h	β	se	t	p	emp. p	n
BoE household focus	0	+0.037	0.150	+0.24	0.809	—	33
BoE household focus	3	−0.038	0.297	−0.13	0.899	—	30
BoE household focus	6	−0.544	0.273	−1.99	0.061 *	0.145	27
BoE household focus	9	−0.330	0.272	−1.22	0.242	—	24
BoE household focus	12	+0.038	0.271	+0.14	0.892	—	21
Press household framing	0	+0.154	0.094	+1.64	0.113	0.139	33
Press household framing	3	+0.165	0.200	+0.82	0.420	—	30
Press household framing	6	−0.195	0.270	−0.72	0.478	—	27
Press household framing	9	−0.619	0.236	−2.62	0.019 **	0.070	24
Press household framing	12	−0.033	0.250	−0.13	0.898	—	21

Table 38: COVID+ (2020–01 to 2026–07)

regressor	h	β	se	t	p	emp. p	n
BoE household focus	0	-0.217	0.193	-1.12	0.278	—	25
BoE household focus	3	+0.112	0.213	+0.53	0.606	—	25
BoE household focus	6	-0.176	0.329	-0.53	0.601	—	23
BoE household focus	9	+0.497	0.336	+1.48	0.161	—	22
BoE household focus	12	-0.305	0.396	-0.77	0.454	—	21
Press household framing	0	+0.204	0.058	+3.52	0.001 ***	0.001	77
Press household framing	3	+0.250	0.111	+2.25	0.028 **	0.042	75
Press household framing	6	+0.165	0.113	+1.45	0.151	—	72
Press household framing	9	+0.059	0.138	+0.43	0.672	—	69
Press household framing	12	-0.026	0.170	-0.16	0.876	—	66

Table 39: COVID onset (2020–2021)

regressor	h	β	se	t	p	emp. p	n
Press household framing	0	+0.261	0.128	+2.04	0.060 *	0.064	23
Press household framing	3	+0.018	0.474	+0.04	0.970	—	20

Table 40: Surge (2021–06 to 2023–12)

regressor	h	β	se	t	p	emp. p	n
Press household framing	0	+0.214	0.095	+2.26	0.034 **	0.042	30
Press household framing	3	+0.167	0.158	+1.06	0.303	—	27
Press household framing	6	+0.193	0.182	+1.06	0.305	—	24
Press household framing	9	+0.220	0.159	+1.38	0.191	—	21
Press household framing	12	+0.010	0.084	+0.11	0.911	—	18

Table 41: Disinflation (2023–2026)

regressor	h	β	se	t	p	emp. p	n
Press household framing	0	+0.067	0.141	+0.48	0.637	—	41
Press household framing	3	+0.091	0.216	+0.42	0.677	—	39
Press household framing	6	-0.141	0.133	-1.06	0.298	—	36
Press household framing	9	-0.201	0.112	-1.79	0.085 *	0.122	33
Press household framing	12	+0.287	0.204	+1.41	0.173	—	30

Table 42: Placebo — does the regressor 'predict' the past?

episode	regressor	3 months before	6 months before
GFC (2007–01 to 2009–12)	BoE household focus	−0.105 (p=0.409)	−0.243 (p=0.258)
GFC (2007–01 to 2009–12)	Press household framing	+0.047 (p=0.699)	+0.143 (p=0.474)
COVID+ (2020–01 to 2026–07)	BoE household focus	−0.124 (p=0.183)	−0.022 (p=0.929)
COVID+ (2020–01 to 2026–07)	Press household framing	+0.050 (p=0.467)	−0.014 (p=0.895)

A3.5 The distribution of household beliefs

Table 43: Household focus and each moment of the distribution

dependent variable	h=2 (6m)	h=4 (12m)	h=6 (18m)	h=8 (24m)
median expectation	−0.106 p=0.289	−0.142 p=0.374	+0.120 p=0.525	+0.222 p=0.269
DISAGREEMENT (Carroll III)	+0.002 p=0.940	−0.039 p=0.203	−0.056 p=0.120	−0.036 p=0.243
share expecting 5%+	−1.353 p=0.260	−2.340 p=0.231	+0.715 p=0.755	+1.957 p=0.465
share expecting <2%	+2.393 p=0.136	+2.742 p=0.295	−1.156 p=0.706	−2.606 p=0.376
share 'don't know'	+0.256 p=0.611	+0.002 p=0.995	+0.580 p=0.392	+0.654 p=0.441
target credibility	−0.430 p=0.652	−0.242 p=0.854	−1.269 p=0.369	−1.049 p=0.511
5-year share 5%+	+0.319 p=0.607	−0.229 p=0.795	+1.323 p=0.227	+1.416 p=0.199
Citi/YouGov 1y	−0.075 p=0.542	−0.062 p=0.750	+0.273 p=0.234	+0.347 p=0.158

Table 44: Same, restricted to quarters when expectations are moving

dependent variable	h=2	h=4	h=6	h=8
median expectation	−0.027 p=0.871	+0.125 p=0.509	+0.357** p=0.012	+0.545*** p=0.001
DISAGREEMENT (Carroll III)	+0.016 p=0.669	−0.002 p=0.936	+0.021 p=0.356	+0.077*** p=0.000
share expecting 5%+	+0.602 p=0.767	+2.692 p=0.234	+5.194*** p=0.007	+7.765*** p=0.001
share expecting <2%	+1.874 p=0.500	+0.059 p=0.984	−4.195 p=0.111	−5.603** p=0.016
share 'don't know'	−0.966 p=0.326	−0.889 p=0.160	−0.778 p=0.223	−2.432*** p=0.000
target credibility	−1.744 p=0.202	−3.424* p=0.066	−4.257** p=0.025	−5.281** p=0.017
5-year share 5%+	−0.363 p=0.604	+0.058 p=0.956	+1.406 p=0.159	+1.779* p=0.077
Citi/YouGov 1y	+0.113 p=0.570	+0.381 p=0.110	+0.772*** p=0.000	+0.959*** p=0.000

Table 45: Does it have to be household focus? The three tone measures compared

tone measure	disagree. $h = 4$	disagree. $h = 6$	median $h = 4$	median $h = 6$
cb_households	-0.039 p=0.203	-0.056 p=0.120	-0.142 p=0.374	+0.120 p=0.525
household focus				
cb_tone	-0.011 p=0.697	+0.004 p=0.861	+0.190 p=0.245	+0.223 p=0.259
sentiment (pos-neg)				
hawk_boe	-0.063** p=0.010	-0.089*** p=0.003	+0.044 p=0.756	+0.118 p=0.506
dictionary				

Table 46: Carroll's second implication — does attention amplify it?

dependent variable	h	interaction β	t	p	n
share 'don't know'	4	-1.589	-1.78	0.080 *	79
share 'don't know'	6	-1.900	-2.29	0.025 **	77

Table 47: The randomisation test and why the nominal p-values must not be quoted

moment	h	t	nominal p	empirical p	verdict
share 'don't know'	8	4.63	0.000	0.017	survives
DISAGREEMENT	8	4.91	0.000	0.040	survives
Citi/YouGov 1y	6	3.99	0.000	0.044	survives
median	8	3.70	0.001	0.066	marginal
share 5%+	8	3.84	0.001	0.069	marginal
median	6	2.66	0.012	0.133	fails
share 5%+	6	2.88	0.007	0.115	fails
target credibility	8	2.51	0.017	0.180	fails

Table 48: What actually survives, and what it means

moment	direction	reading
share saying "don't know"	falls	households form an opinion where they previously had none
disagreement	rises	but they do not converge on the same opinion
level (Citi/YouGov)	rises	and the centre drifts up

A3.6 Identified shocks against the distribution

Table 49: Press household framing (reduced form) (**s_news**) — 59 quarters available

dependent variable	h=0	h=2	h=4	h=6	h=8	n
median expectation	+0.047	+0.105	−0.233	−0.307*	−0.348*	47
mean expectation	+0.043	+0.071	−0.169	−0.196	−0.261	47
share expecting 5%+	+0.654	+1.299	−2.658	−3.517*	−4.614**	47
share expecting <2%	−0.814	−1.248	+2.894	+3.632	+4.725	47
share 'don't know'	+1.647**	+0.030	+0.478	+2.040***	+1.692***	47
disagreement (sd)	−0.014	−0.004	−0.022	−0.038	−0.026	47
target credibility	−0.967*	+0.106	+2.196*	+1.596	+2.646**	47
5-year share 5%+	+0.826	+0.096	−0.895	−1.284	−0.116	33
Citi/YouGov (benchmark)	+0.036	−0.160	−0.443*	−0.465*	−0.553***	47

Table 50: BoE household framing (reduced form) (**s_boe**) — 86 quarters available

dependent variable	h=0	h=2	h=4	h=6	h=8	n
median expectation	−0.084	−0.108	−0.266**	−0.056	+0.107	77
mean expectation	−0.061	−0.120	−0.223**	−0.056	+0.072	77
share expecting 5%+	−0.405	−0.906	−3.161**	−0.969	+1.169	77
share expecting <2%	+1.611*	+2.785**	+4.207**	+1.356	−0.978	77
share 'don't know'	−0.587	+0.048	−0.184	+0.413	+0.652	77
disagreement (sd)	+0.018	+0.037	−0.002	−0.024	−0.036	77
target credibility	−0.522	−0.406	+0.270	−0.991	−1.864	77
5-year share 5%+	−0.375	+0.180	−1.296	+0.569	+0.262	60
Citi/YouGov (benchmark)	+0.011	−0.030	−0.178	+0.101	+0.282	74

Table 51: Information shock (**eps_cbi**) — 86 quarters available

dependent variable	h=0	h=2	h=4	h=6	h=8	n
median expectation	−0.008	−0.017	+0.019	+0.014	−0.009	76
mean expectation	−0.002	−0.012	+0.009	+0.002	−0.016	76
share expecting 5%+	−0.182	−0.202	+0.145	+0.236	+0.485	76
share expecting <2%	−0.075	+0.183	−0.190	+0.103	+0.736	76
share 'don't know'	−0.216	−0.147	+0.245	−0.139	+0.099	76
disagreement (sd)	−0.010	−0.007	−0.003	−0.002	+0.014	76
target credibility	+0.220	+0.036	−0.092	−0.051	−0.224	76
5-year share 5%+	−0.398**	−0.355	−0.116	−0.293	−0.106	60
Citi/YouGov (benchmark)	+0.026	+0.034	+0.072	+0.065	+0.022	74

Table 52: Stance shock (`eps_mp`) — 86 quarters available

dependent variable	h=0	h=2	h=4	h=6	h=8	n
median expectation	-0.012	-0.037	-0.119**	-0.091	-0.063	76
mean expectation	-0.011	-0.022	-0.098**	-0.078*	-0.062	76
share expecting 5%+	+0.215	-0.300	-1.396**	-1.065	-0.670	76
share expecting <2%	+0.258	+0.242	+1.742**	+1.265	+1.232	76
share 'don't know'	-0.312	-0.134	-0.012	+0.029	+0.566	76
disagreement (sd)	+0.012	+0.011	+0.012	+0.007	+0.014	76
target credibility	+0.206	+0.189	+0.353	+0.391	+0.118	76
5-year share 5%+	-0.255	-0.315	-1.139***	-0.695*	-0.363	60
Citi/YouGov (benchmark)	-0.007	+0.020	-0.075	-0.035	-0.038	74

Table 53: Romer-Romer policy shock (`policy_shock_std`) — 42 quarters available

dependent variable	h=0	h=2	h=4	h=6	h=8	n
median expectation	-0.050	-0.093	-0.118	+0.068	+0.041	28
mean expectation	-0.060*	-0.076	-0.071	+0.077	+0.017	28
share expecting 5%+	-0.273	-1.298	-1.652	+0.410	-0.780	28
share expecting <2%	+1.955**	+0.811	+0.798	-1.301	-0.798	28
share 'don't know'	-0.108	-0.725**	+0.603	-0.741	+1.564	28
disagreement (sd)	-0.004	-0.027*	-0.039**	-0.038	-0.086**	28
target credibility	+0.037	+0.544	+0.027	-0.157	+0.399	28
5-year share 5%+	-0.751**	-0.087	-0.011	+0.739	+0.896	28
Citi/YouGov (benchmark)	-0.063*	-0.058	+0.058	-0.036	+0.027	29

A3.7 Transmitted and autonomous press framing

Table 54: 1. How much do they actually move together?

pair	corr (levels)	corr (changes)	n
BoE framing vs press framing (keyword)	-0.007	+0.130	146
BoE framing vs press framing (RoBERTa)	+0.163	+0.131	146
press keyword vs press RoBERTa	+0.794	+0.739	173
BoE SURPRISE vs press SURPRISE	-0.003	+0.067	142

Table 55: 2. Who leads whom?

k (months)	levels	surprises	
-12	-0.174	-0.069	
-11	-0.172	-0.103	
-10	-0.208	-0.155	
-9	-0.212	-0.155	
-8	-0.233	-0.157	
-7	-0.129	-0.053	
-6	-0.105	-0.064	
-5	-0.118	-0.079	
-4	-0.115	-0.041	
-3	-0.065	-0.022	
-2	-0.120	-0.079	
-1	-0.095	-0.089	
+0	-0.007	-0.003	
+1	-0.033	+0.014	
+2	-0.054	-0.037	
+3	-0.022	-0.029	
+4	-0.007	-0.021	<- peak
+5	-0.074	-0.057	
+6	-0.080	-0.079	
+7	-0.112	-0.099	
+8	-0.118	-0.126	
+9	-0.100	-0.124	
+10	-0.099	-0.119	
+11	-0.094	-0.107	
+12	-0.212	-0.218	

Table 56: 3. Lead-lag regressions (6 lags, own lags controlled, HC1)

direction	F	p	sum of coefficients	n
BoE framing -> press framing	1.00	0.432	+0.108	71
press framing -> BoE framing	1.54	0.180	-0.056	74
BoE surprise -> press surprise	1.31	0.274	-0.070	54
press surprise -> BoE surprise	1.64	0.159	+0.139	57

Table 57: Local projections on each component, run separately

h	transmitted b	p	autonomous b	p
0	−0.001	0.967	+0.060	0.065
3	−0.067	0.426	+0.053	0.432
6	−0.234	0.023	−0.092	0.204
9	−0.347	0.000	−0.193	0.017
12	−0.300	0.005	−0.198	0.027
18	−0.110	0.265	−0.104	0.373

Table 58: Both components entered in the same regression

h	transmitted b	p	autonomous b	p	n
0	−0.024	0.497	+0.066	0.054	79
3	−0.093	0.283	+0.100	0.164	79
6	−0.212	0.048	−0.014	0.833	79
9	−0.308	0.003	−0.098	0.193	79
12	−0.253	0.027	−0.128	0.168	79
18	−0.066	0.539	−0.086	0.476	79

A3.8 High-frequency event study

Table 59: The clean experiment: pre–2015 minutes releases

tone measure	maturity	β (bp)	se	t	p	n
dictionary	2.5y	−0.518	1.224	−0.42	0.673	91
dictionary	3.0y	−0.648	0.990	−0.65	0.514	109
dictionary	5.0y	−0.433	0.744	−0.58	0.562	122
CB-RoBERTa	2.5y	+2.278	0.969	+2.35	0.021 **	91
CB-RoBERTa	3.0y	+1.994	0.845	+2.36	0.020 **	109
CB-RoBERTa	5.0y	+1.376	0.642	+2.14	0.034 **	122

Table 60: Post–2015 (decision, minutes and MPR released together)

tone measure	maturity	β (bp)	se	t	p	n
dictionary	2.5y	−1.196	0.785	−1.52	0.130	113
dictionary	3.0y	−1.029	0.661	−1.56	0.122	131
dictionary	5.0y	−0.886	0.457	−1.94	0.055 *	139
CB-RoBERTa	2.5y	+0.751	1.195	+0.63	0.531	113
CB-RoBERTa	3.0y	+0.011	1.006	+0.01	0.991	131
CB-RoBERTa	5.0y	+0.030	0.641	+0.05	0.963	139

Table 61: Window length

sample	tone	window	β (bp)	t	p	n
pre-2015 minutes	dictionary	$[-1,0]$	-0.191	-0.32	0.749	109
pre-2015 minutes	dictionary	$[-1,+1]$	-0.648	-0.65	0.514	109
pre-2015 minutes	dictionary	$[-1,+5]$	-0.618	-0.28	0.779	108
pre-2015 minutes	CB-RoBERTa	$[-1,0]$	+1.195	+1.79	0.077 *	109
pre-2015 minutes	CB-RoBERTa	$[-1,+1]$	+1.994	+2.36	0.020 **	109
pre-2015 minutes	CB-RoBERTa	$[-1,+5]$	+2.882	+1.49	0.140	108
post-2015	dictionary	$[-1,0]$	-0.565	-0.96	0.338	131
post-2015	dictionary	$[-1,+1]$	-1.029	-1.56	0.122	131
post-2015	dictionary	$[-1,+5]$	-2.105	-1.51	0.135	131
post-2015	CB-RoBERTa	$[-1,0]$	+0.241	+0.34	0.735	131
post-2015	CB-RoBERTa	$[-1,+1]$	+0.011	+0.01	0.991	131
post-2015	CB-RoBERTa	$[-1,+5]$	+0.258	+0.13	0.893	131

Table 62: Placebo and pre-trend — windows before and around the event

window	β (bp)	t	p	n
$[-6,-4]$ BEFORE (placebo)	+0.322	+0.72	0.473	108
$[-4,-2]$ BEFORE (placebo)	+1.437	+2.19	0.031 **	109
$[-1, 0]$ event	+1.195	+1.79	0.077 *	109
$[-1,+1]$ event	+1.994	+2.36	0.020 **	109
$[-1,+3]$	+1.852	+1.36	0.176	109
$[-1,+10]$	+2.268	+0.76	0.451	108

Table 63: Placebo and pre-trend — controlling for the pre-window move directly

specification	β (bp)	t	p	n
baseline	+2.011	+2.37	0.020 **	107
+ 5-day pre-window move	+1.220	+1.48	0.141	107
+ 10-day pre-window move	+1.708	+2.19	0.031 **	107
+ both	+1.210	+1.46	0.146	107

A3.9 Reported nulls re-estimated

Null 6 is the term structure of inflation compensation, which Section 6.3 reports as evidence that the measure carries information markets act on, and Null 10 is the monthly version of the market test, retained to document that the monthly frequency (the daily event study in A3.8 is the clean version). The third, Null 5, is a single marginal coefficient on the type-token ratio at eighteen months out of twelve estimated across three readability measures, which is what one result out of twelve looks like under a 10% threshold. Nulls 7 and 8 return nothing.

Every test here was originally estimated with Newey-West standard errors, which Section 5.3 rejects because they over-reject at long horizons, and each is re-run on the current specification: lag-augmented local projections with HC1 errors. The attention interaction, originally listed as Null 3, is reported in Section 6.4 and in A3.10 instead, on the specification the randomisation test was run against.

Table 64: Null 5 — complexity and readability

regressor	h=6	h=9	h=12	h=18	n
Gunning fog index	+0.019	−0.060	−0.073	−0.055	156
Flesch–Kincaid grade	+0.018	−0.045	−0.065	−0.081	156
type–token ratio	+0.030	−0.024	−0.088	−0.201**	156

Table 65: Null 6 — term structure of inflation compensation

dependent variable	h=6	h=9	h=12	h=18	n
3-year	−0.191**	−0.295***	−0.261***	−0.181**	137
5-year	−0.151***	−0.233***	−0.193***	−0.141**	143
10-year	−0.109***	−0.165***	−0.142***	−0.090**	143
15-year	−0.070**	−0.111***	−0.099***	−0.060*	143
20-year	−0.035	−0.061**	−0.049**	−0.020	143

Table 66: Null 7 — MPC dissent as a regressor

regressor	h=6	h=9	h=12	h=18	n
dissent share (orthogonalised)	+0.119*	+0.088	+0.046	−0.028	156

Table 67: Null 8 — cumulative (three-month) framing surprise

regressor	h=6	h=9	h=12	h=18	n
press framing, single month (benchmark)	−0.128*	−0.284***	−0.346***	−0.378***	143
press framing, 3-month cumulative	−0.032	−0.264	−0.247	−0.316	131

Table 68: Null 10 — monthly market tests: Dictionary hawkishness

dependent variable	h=6	h=9	h=12	h=18	n
1-year swap	+0.142	+0.127	+0.280***	+0.398***	141
5-year swap	+0.017	+0.048	+0.063	+0.064	141
5-year breakeven	+0.068	+0.108	+0.202***	+0.111*	62
5y5y forward	−0.040	−0.035	−0.058**	−0.052*	165

Table 69: Null 10 — monthly market tests: RoBERTa tone

dependent variable	h=6	h=9	h=12	h=18	n
1-year swap	+0.226**	+0.182	+0.166	+0.085	141
5-year swap	+0.114***	+0.078	+0.099*	+0.065	141
5-year breakeven	+0.088	+0.120	+0.170*	+0.134**	62
5y5y forward	+0.067**	+0.029	+0.057*	+0.030	165

A3.10 Diagnostics

Table 70: Empirical size of the local projection under two estimators

horizon	Newey–West, no lag augmentation	lag-augmented, HC1	
h = 6	8.3%	7.4%	
h = 9	10.5%	6.8%	<–
h = 12	12.0%	6.2%	<–
h = 18	15.3%	7.5%	<–

Table 71: Randomisation test on the attention interaction

h	real interaction	analytical p	empirical p	n
6	+0.125	0.105	0.173	155
9	+0.198	0.015	0.055	152
12	+0.281	0.000	0.006	149
18	+0.288	0.000	0.008	143

A3.11 LASSO selection

Selection over the candidate set at the twelve-month horizon, reported in Section 6.2. Candidates observed for less than 80% of the panel are dropped. Forty-five candidates enter over 114 complete months, June 2007 to June 2025. LASSO coefficients are shrunk towards zero and carry no standard errors, so the tables report selection and not inference.

Table 72: Five-fold random CV

variable	LASSO coefficient
brent_usd_bbl	-0.4810
cpi_food_yoy	-0.4733
news_hh_focus	-0.1940
cb_households	-0.1527
swap_1y	+0.1511
cb_positive	+0.1460
cb_firms	+0.1436
cpi_energy_yoy	-0.0975
cpi_yy_surprise	+0.0874
minutes_ttr	-0.0867
gt_cost_of_living	-0.0746
gfk_consumer_confidence	+0.0441

Table 73: Blocked time-series CV

variable	LASSO coefficient
cpi_food_yoy	-0.4069
brent_usd_bbl	-0.2725
news_hh_focus	-0.1844

Table 74: Penalty path

alpha	survivors	framing retained
0.05	17	yes
0.10	12	yes
0.15	8	yes
0.20	5	yes
0.30	3	yes
0.40	3	yes
0.50	3	yes
0.70	3	yes

Table 75: Seed sensitivity, five-fold random CV

seed	alpha	survivors	framing retained
0	0.0731	12	yes
1	0.0223	24	yes
2	0.0169	25	yes
3	0.0169	25	yes
20260828	0.0111	29	yes

Table 76: Diagnostic: the cost of retaining a thinly observed candidate. **hawk_news**, the press hawkishness dictionary, is observed for 47% of the panel. Because selection runs on complete cases, retaining it removes a third of the estimation sample, and on the shorter sample the framing measure is no longer selected.

candidate set	complete months	survivors	framing retained
without hawk_news	114	12	yes
with hawk_news forced in	77	8	no